The Effects of Body-worn Cameras on Policing and Court Outcomes: Evidence from Virginia Criminal Courts

Katie Bollman

Oregon State University

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Abstract: Body-worn cameras monitor police interactions and provide a new source of evidence for court cases, but do so at a cost of attorney time. Using data on criminal charges and the staggered adoption of body-worn cameras in Virginia state court jurisdictions from 2009-2019, I test whether body-worn cameras affect policing and court outcomes. I find that adoption reduced new case filings for offenses initiated during a police interaction, like resisting arrest, by nearly ten percent, suggesting improved interactions between police and the public. However, I do not find evidence that body-worn cameras changed case outcomes on average, nor do I find evidence of a disparate impact for Black and non-Black defendants.

1 Introduction

Defining and implementing effective policing remains one of the most salient political issues of the past decade. In the midst of rigorous debate over policing policies in the United States, outfitting law enforcement with body-worn cameras (BWCs) has broad public support.[1] Between 2020 and mid-2021, six states mandated body-worn cameras for law enforcement (NCSL, 2021).

Although increasingly commonplace, body-worn cameras are a recent technological advancement for law enforcement in the United States. In 2010, less than 5 percent of law enforcement agencies used body-worn cameras (Bureau of Justice Statistics, 2016). This changed rapidly after a police officer killed eighteen-year-old Michael Brown in Ferguson, Missouri in August 2014 (Buchanan et al., 2015), leading Ferguson police to adopt body-worn cameras to increase transparency, accountability, and public trust (BBC, 2014). By late 2016, nearly half of U.S. general purpose law enforcement agencies used body-worn cameras (Bureau of Justice Statistics, 2016).[2]

Body-worn camera advocates intend the technology to increase transparency and improve safety in police interactions, reducing misconduct and some visible criminal activities; criminal justice practitioners describe broader effects on policing and the courts. In addition to inducing behavioral changes among police and the public, body-worn cameras generate recorded data that may be relevant to criminal cases. While these data may allow for a more accurate resolution of criminal charges, accessing the evidentiary value of the data comes at a cost of attorney time. Across Virginia, attorneys report that the tension between attorney time constraints and the additional labor demands of cases with body-worn camera footage can be detrimental for vulnerable criminal defendants. As the Executive Director of the Virginia Indigent Defense Commission wrote, "...we have significant concerns that our attorneys will not be able to continue to meet their ethical and professional responsibilities" (Compensation Board, 2018).

In this paper, I extend the base of research on body-worn cameras as a policing tool to incorporate their effects on courts and defendant outcomes. Specifically, I use the timing of body-worn camera adoption by local law enforcement agencies across the Commonwealth of Virginia to study changes in criminal case filings, resolutions, and court processes. In doing so, I discuss three channels through which body-worn cameras can affect the courts. First, body-worn cameras can induce behavioral effects among law enforcement and the public, which can change the set of cases filed in the courts. Then, once cases are filed, body-worn camera video can introduce evidentiary and attorney time use effects that change the outcomes of the case or the process by which these outcomes are realized.

My contributions build upon work by criminologists (Ariel et al., 2015, Katz et al., 2014, Yokum et al., 2017) and contribute to an emerging literature in economics (Ferrazares, 2024, Kang, 2023, Kim, 2025, Çubukçu et al., 2021). More broadly, by exploring behavioral effects I contribute to literatures on police responses to oversight (Ba and Rivera, 2019), criminal responses to surveillance (Gonzalez-Navarro, 2013, Gómez et al., 2020, Piza et al., 2019), and

[1] One 2020 poll showed 85 percent of Republicans and 94 percent of Democrats favor body-worn camera mandates (Kull, 2020)

[2] In 2015 the Department of Justice announced a $75 million national grant program intended to fund 50,000 cameras over a three-year period (Department of Justice, Office of Public Affairs, 2015).

criminal deterrence (Chalfin and McCrary, 2017). While previous body-worn camera studies often focus on changes in police use-of-force — an important but uncommon outcome — I test for behavioral effects in more common interactions by measuring changes in the overall frequency and composition of new charges.

Less is known about if and how body-worn cameras affect court processes and case resolutions once charges are filed. Two local impact evaluations provide contradicting evidence: Yokum et al. (2017) do not find effects of body-worn cameras on case outcomes in Washington D.C., whereas Katz et al. (2014) note some prosecutorial changes coincided with body-worn camera adoption in Phoenix. Outside of criminal courts, Çubukçu et al. (2021) find that body-worn camera footage evidence affects the resolution of citizen complaints against police. I provide some of the first empirical evidence around these court-based effects and the first evidence using data from multiple court jurisdictions.

To do so, I collected a new data set detailing the timing of body-worn camera adoption across Virginia court jurisdictions. Existing data collections on body-worn camera adoption suffered from poor data quality in key fields or employed sampling structures best suited for agency-level analyses. Because multiple law enforcement agencies can operate within a single court jurisdiction, court-level analyses with a sampling of agency-level data would be limited to those courts for which all key agencies in the court jurisdiction were sampled. These data constraints led to a high representation of urban areas in body-worn camera evaluations and less evidence from small and mid-sized localities. In contrast, my adoption data reflects the broadest coverage of Virginia law enforcement agencies to date, covering the major law enforcement agencies in 90 percent of Virginia court jurisdictions. I combine these new body-worn camera data with a second data set containing the near-universe of criminal court charges in Virginia from 2009-2019 to create court-level panels covering 101 Virginia circuit courts and 108 Virginia district courts. I analyze these data in a difference-in-differences framework. Because law enforcement began using body-worn cameras at different points in time, I implement the Borusyak et al. (2021) imputation estimator (BJS) to test for effects of body-worn cameras on case filings, processes, and resolutions.

Across two court levels and several outcomes measuring changes in case filings, processes, and resolutions, I find a strong pattern of results. Body-worn cameras reduced the prevalence of a subset of charges arising out of interactions with police including resisting arrest, assaulting an officer, and similar offenses. However, beyond this subset, they do not appear to have meaningfully altered police behaviors as a whole: I do not find significant changes in the number of cases filed, and both the share of those cases involving multiple charges, and the distribution of charge severity only nominally changed after police began using body-worn cameras.

After charges are filed, cases also do not systematically proceed or resolve differently due to body-worn camera adoption — despite practitioner reports of an indigent defense system buckling under the weight of new body-worn camera data. One potential explanation for these null results could be that the evidentiary effects of the videos offset attorney time use effects for a net zero effect. However, I show that there are no differential effects of body-worn cameras on cases more likely to have associated video. Because a common pool of attorneys litigate video and non-video cases, if attorneys substitute time across the cases they represent then the lack of a differential effect suggests that the offsetting-effects hypothesis does not hold. Finally, because the increase in body-worn camera programs across the U.S. was motivated in part by racial disparities in policing, I test for evidence of differential effects for Black and non-Black defendants. I do not find compelling evidence that body-worn cameras differentially helped or harmed Black defendants.

The paper proceeds as follows. Section 2 provides an institutional context for Virginia courts and body-worn camera adoption, and an economic context for understanding the consequences of court outcomes. Section 3 presents a conceptual framework for the three primary channels through which body-worn cameras may affect court cases: behavioral effects, evidentiary effects, and attorney time use effects. Section 4 contains a description of the court and body-worn camera adoption data that I use for my court-level analyses; Section 5 outlines the empirical strategies that I use to analyze these data. Section 6 presents aggregate and temporally decomposed results. Section 7 dissects the null results and tests for heterogeneity in treatment effects by the race of the defendant and for cases more likely to include body-worn camera video. Section 8 concludes.

2 Background

2.1 Body-worn cameras

Body-worn cameras became a commonplace tool for U.S. law enforcement agencies in the latter half of the 2010s. A 2016 national survey of nearly 4,000 U.S. law enforcement agencies documented both a rapid expansion in use between 2013 and 2016, as well as high intentions among non-adopters to use the technology in the future. Thirty-one percent of non-adopting agencies reported that they were likely or very likely to consider acquiring body-worn cameras in the following year. Even agencies that did not intend to imminently adopt reported high rates of officer and community support for such programs (Bureau of Justice Statistics, 2016). Agencies adopted body-worn cameras to improve officer safety (21 percent) and accountability (19 percent), and to reduce or more quickly resolve citizen complaints (15 percent).[3] While evidentiary benefits rarely drove adoptions, agencies were optimistic that the videos would make cases more prosecutable (69.8 percent) and improve evidence quality (78.8 percent).

Non- or not-yet adopting agencies most often cited financial constraints – a common obstacle to implementation for many adopting agencies as well. To achieve the expected policing benefits, adopting agencies almost always established a formal policy outlining body-worn camera use.[4] Policies typically dictate that cameras be activated for traffic stops (93 percent), when executing arrest or search warrants, deploying firearms, and initiating contact with the public (nearly 85 percent each), and be preserved for between one month and one year. Retention periods are often extended if pertinent to an ongoing matter, such as a use of force incident, citizen complaint, or if used as evidence in a legal proceeding.

I show in Figure 1 that Virginia law enforcement followed a similar adoption trend to the U.S. overall: in the LEMAS-BWCS survey, sixty-two percent of responding Virginia agencies adopted body-worn cameras by late 2016. Using new collected data, described in section 4,

[3]Percentages exclude respondents whose agency's primary purpose was to conduct a pilot program.

[4]Jordan et al. (2026) shows evidence of imperfect compliance with camera policies – aligned with this, my study evaluates the effects of body-worn camera programs as they are practically implemented.

I also show that pace of adoptions tapered after the 2015 peak, but departments continued to routinely adopt body-worn cameras through 2018.

2.2 Virginia Courts

Body-worn camera videos recorded during police-public interactions generate large volumes of video data in part because police interact with the public often. The Bureau of Justice Statistics estimates that in 2018, about 24 percent of the U.S. population (60 million people) interacted with the police (Harrell and Davis, 2020), although many of these contacts were resident-initiated and did not result in criminal charges. In 2013, Virginia State General District and Circuit Courts received a total of nearly 900,000 felony and misdemeanor filings (OES, 2014a,b).

These two sets of courts, district and circuit level, serve as the primary venues for criminal litigation across Virginia. The courts largely share geographic jurisdictions, with approximately one circuit court and one district court in each county or independent city across the state. However, they differ in the scope of the cases they hear: district courts hold jurisdiction over misdemeanor cases and preliminary hearings for felony cases, whereas the circuit courts litigate felonies. Courts often receive cases from multiple law enforcement agencies within their geographic jurisdictions: for example, both a county sheriff and a town police department.

Charges are filed in the district courts through two primary mechanisms: arrests or summonses. Summons are used for many misdemeanors and do not require the defendant be held in custody while awaiting court proceedings. Alternatively, police can arrest defendants. When this happens, a local magistrate provides an intermediate step between law enforcement and the courts. Magistrates review sworn statements from a complainant (such as an arresting officer) to assess whether there is probable cause to proceed with a criminal charge. This standard of probable cause – that "the charges are not capricious and are sufficiently supported to justify bringing into play the further steps of the criminal process" (Department of Magistrate Services, Office of the Executive Secretary, 2021) is much weaker than a standard to convict.

After a summons is issued or the charges advance from the magistrate's office, a judge, prosecutor, and defense attorney can each affect the outcome of the charges.[5] I provide a basic case road map outlining the entities involved in various states of criminal litigation in Table 1. The court actors can influence defendant outcomes ranging from the final set of charges he will face to the outcomes of those charges, their sentencing, and even the pace at which the case resolves. More specifically, prosecutors can alter, drop, or add charges to a criminal case; judges dismiss or rule on charges and determine sentences; and both prosecuting and defense attorneys lobby for preferred outcomes (dispositions) and sentencing.

A key motivation for this paper is the reports from prosecutors and indigent defenders warning that the marginal time required to review body-worn camera footage exceeds attorney time constraints. On paper, even prior to body-worn camera adoption, full-time indigent

[5]Statutorily, all three actors are involved in felony cases. Practically, all three actors are involved in many misdemeanor cases as well. However, for low-level misdemeanors and infractions a prosecutor and/or defense attorney may not be involved in the case.

defenders in Virginia were likely to face binding time constraints for their caseloads.[6]

3 Literature and Conceptual Framework

3.1 Behavioral Effects

"That's the beauty of these devices . . . everybody gets politer when the cameras are on." -Norfolk Police Chief Michael Goldsmith

As an evidence-generating technology, body-worn cameras reduce the noise around allegations of criminal behavior or professional misconduct. The most publicized examples involve cases of excessive force by police. However, offenses committed by members of the public that are caught on camera may also be easier to prosecute. Consequently, the mere presence of a body-worn camera may be sufficient to change court outcomes by bringing police and public behavior into alignment with legal and social standards. Prior studies show new technologies that increase the expected costs to criminal activities reduce those behaviors (Ayres and Levitt, 1998, DiTella and Schargrodsky, 2013, Doleac, 2017, Gonzalez-Navarro, 2013). Criminologists show closed-circuit television (CCTV), the technology perhaps most similar to body-worn cameras, also reduces crime by approximately 13 percent (Piza et al., 2019). Notably, CCTV and body-worn camera technologies both create video evidence, but differ in their applications; whereas CCTV can be paired with remote monitoring to enable real-time responses, because body-worn cameras are worn by officers (monitors) this technology has no additional direct monitoring effects. Even so, the crime-reducing effects of CCTV are not solely attributable to active monitoring (Gómez et al., 2020), suggesting that recorded video from body-worn cameras may similarly exert a deterrent effect.

For police, body-worn cameras enable additional oversight, which may improve conduct among officers and sort low-quality officers from the ranks. Criminologists show mixed results on police use-of-force and complaint outcomes across various randomized controlled trials (Ariel et al., 2015, 2016, Yokum et al., 2017). Economists expand this literature with non-experimental methods showing reduced use-of-force by police (Ferrazares, 2024, Kim, 2025).

Although use-of-force and excessive force are socially important outcomes to study, these events are relatively rare in policing; for example, 84 percent of Chicago officers generated no use-of-force complaints over a 5-year period (Chalfin and Kaplan, 2021). Conceptually, behavioral effects of body-worn cameras may also appear in more common events. For example, more deferential defendants and officers may reduce the frequency of charges of officer-oriented offenses such as resisting arrest. And officers, who have a degree of discretion in issuing citations and making arrests, may be less likely to overcharge criminal defendants.

[6] In FY 07/08, before body-worn cameras became widespread, salaried public defenders in Virginia managed on average 320 cases per attorney per year (Kleiman and Lee, 2010). Online appendix D shows that these caseloads exceed the American Bar Association's recommendation of a maximum of 150 felonies *or* 400 misdemeanors (Association, 2009) annually. Similarly, assigned counsel, publicly funded attorneys paid on a case-by-case basis, face strict compensation caps that amount to approximately 1.3 hours of paid work on misdemeanor charges at the district court level, and less than 5 hours of paid work on a typical circuit court felony charge (Code of Virginia 19.2-163).

However, they also may be disincentivized from displaying leniency if they anticipate that their footage will be reviewed. These alterations could affect defendants on both the intensive and extensive margins, reducing the probability an individual is accused of a first offense or that they are charged with multiple offenses stemming from a single incident. Evidence toward these dynamics is mixed; Kang (2023) finds increases in investigatory stops following body-worn camera adoption in New Orleans, whereas Ferrazares (2024) shows evidence of de-policing in Chicago.

3.2 Evidentiary Effects

After charges are filed, body-worn camera video provides evidence that may affect how judges and juries perceive the events that unfolded during a police interaction. Because many cases resolve through plea negotiations, case outcomes depend not only these realized perceptions, but how defendants and attorneys predict the evidence will be perceived. Rational plea negotiations will take into account the probability and severity of conviction/sentencing (Butcher et al., 2021). Additional evidence can influence these plea negotiations by improving the bargaining position of one side. In the case of body-worn cameras, video may reveal law enforcement error or abuse or may corroborate/undermine defendant or law enforcement accounts of events.

Ultimately, the extent and balance of evidentiary value is an empirical question with limited research to date. Officers believe body-worn cameras make cases more prosecutable (Katz et al., 2014), and cameras demonstrably increase disciplinary actions against officers due to substantiated complaints Çubukçu et al. (2021) –affirming the evidentiary value of video, at least in some cases.

3.3 Time Use

"It's a razor thin wire, because you're looking to be sure your client's due process rights are preserved. On the other hand, I have 120 other clients. I have to preserve their due process rights too."

-Newport News Public Defender Robert Moody (Albiges, 2019)

Ascertaining whether body-worn camera video provides evidentiary value to a case requires a review of available footage, of which there can be large quantities. In a 2018 survey of Virginia prosecutors, 51 offices estimated that they received a total of 180,000 hours of body-worn camera footage over a 12 month period (Compensation Board, 2018). Similarly, three Virginia public defender offices reported spending between 160 and nearly 3000 hours per month on body-worn camera related tasks, the workload of between one and 16 additional full-time employees in offices staffed by fewer than ten attorneys attorneys (Gaub et al., 2019). However, staffing levels for publicly provided attorneys are sticky, and assigned counsel compensation caps depend only on case types — not on the workload or available evidence. One hour of video review time (the average reported by Henrico County prosecutors for cases with video (Compensation Board, 2018)) amounts to over 20% of the total time for which assigned counsel are compensated on typical felony charge.

Some of the time, body-worn camera video review may substitute for other case tasks. Other times, attorney time constraints may cause this review to crowd out non-body-worn camera activities. The extent to which each takes place remains ambiguous due to limited data. However, practitioners report crowd-out: for example, the Executive Director of the Virginia Indigent Defense Commission wrote, "it is not hard to imagine that court-appointed attorneys will be faced with terrible choices, which will hurt their clients, hurt their practice, or potentially undermine both. Court-appointed attorneys will likely have to stop taking court-appointed cases; not watch all the body-worn camera footage, in violation of their ethical duties; or basically be forced to work for free" (Compensation Board, 2018). The Ethics Counsel for the Virginia State Bar echoed this sentiment on behalf of prosecutors, stating "Existing prosecutors' workloads will be significantly increased by the time taken to review footage derived from body-worn cameras. To comply with legal and ethical standards, Commonwealth's Attorneys must staff more lawyers or decline handling cases. Breaching the legal and ethical standards is obviously not an option" (Compensation Board, 2018). These practitioner concerns subsequently led to policy efforts in Virginia aimed at ameliorating attorney time use effects (Virginia Association of Counties, 2019).

4 Data

4.1 Treatment Data

Existing data on body-worn cameras have limited coverage of law enforcement agencies and court jurisdictions. I fill this gap by collecting a more comprehensive set of law enforcement body-worn camera adoption data in Virginia, which I aggregate to a quarterly court-level adoption indicator. Observing body-worn camera adoption at the court-level rather than the case-level allows me to take a broad view of direct and spillover effects of body-worn cameras on cases with and without footage. I combine these body-worn camera data with charge-level data from Virginia courts to form quarterly court-level panels that I use to explore changes in charging, case processes, and case resolutions.

I measure body-worn camera adoption at the geographic court jurisdiction level; courts are "treated" when the first major policing agency operating in their jurisdiction implements a body-worn camera program. Multiple agencies of varying sizes may operate within a single jurisdiction, so I use the "major" designation to focus on those agencies likely to contribute influentially to court caseloads. For this same reason, I also exclude agencies without policing duties or with primary duties related to jail and court security. Among the remaining eligible agencies, I using the 2016 Law Enforcement Agency Roster (United States Department of Justice, 2017) to define an agency as "major" if it employed at least 25 percent of the total officers or served at least 25 percent of the population in its court jurisdiction.[7] I sought information about body-worn camera adoption for these agencies through Freedom of Information Act (FOIA) requests. I extended FOIA requests to 157 qualifying agencies and obtained information from an additional 32 agencies through direct contact, departmental websites, and local media. Ultimately I obtained adoption data for 111 district court jurisdictions including 78 that adopted body-worn cameras by 2019 and

[7] I show the robustness to an alternative 50% threshold in C.

104 circuit court jurisdictions, 76 of which adopted by 2019. These comprise nearly 90 percent of state district and circuit courts in Virginia.

4.2 Sample Construction and Outcomes

I combine the body-worn camera adoption data with administrative records on charges filed in Virginia district and circuit courts between January 2009 and March 2019, obtained from the Virginia Court Data repository (Schoenfeld, 2021). Concluding in March 2019 allows cases one year to resolve prior to the onset of the COVID-19 pandemic and resulting court shutdowns. For a balanced panel with two years of pre- and one year of post-adoption data for all adopting localities, I drop localities that adopted body-worn cameras before 2011 or after the first quarter of 2018. I present supplemental results with the unbalanced panel in Appendix C.

Data include defendant characteristics (race and sex), charge filing date, disposition, sentence, and offense descriptions. Charges most often conclude in a guilty disposition, charge dropped by the prosecutor, or judicial dismissal. I also observe whether a charge is amended (superceded by a corrected or alternative charge) after filing. When a defendant faces multiple charges at the same time it is likely that charge characteristics, court processes, and outcomes of the individual charges are interrelated. Thus, I aggregate to the case level when defining defendant outcomes in a process detailed in Appendix A. Once aggregated, I define disposition and sentencing outcomes using binary indicators – for example, whether any charge in the case produced a guilty disposition or resulted in incarceration.[8]

Because the channels for body-worn camera effects occur at distinct times, I use separate case samples to test for behavioral effects in case filings, and evidentiary/time use effects in case processes and resolutions. To evaluate changes in case filings, I create a quarterly court-level panel summarizing all infraction, misdemeanor, and felony filings in district courts. Including all case types captures police interactions that involve the courts without drawing distinctions across the severity of the cases. I create a parallel panel for circuit court case filings, but note that because cases filter through the district court to reach the circuit court, changes in this set of filings may reflect any of the effect channels.

Primary outcomes for both filing panels include the total number of filings per 100,000 residents and the share of cases that include multiple charges – an indicator of case complexity.[9] In district court filings, I also test for changes in the distribution of cases by case type. Another key outcome in district courts is prevalence of a subset of cases likely to be most responsive to behavioral effects. This subset of "behavioral effect charges" (also measured per 100,000 residents) originate or escalate in the presence of a police officer and include disorderly conduct, eluding police, resisting arrest, and assault or other offenses identified as directed toward law enforcement. Moderating behavioral effects should reduce both the number of cases entering the courts and this subset, although the total case count will only show reductions if behavioral effects are widespread.

[8]Sentence length is reported inconsistently by Virginia courts when defendants are convicted on multiple charges, necessitating an alternative summary measure.

[9]I construct per-capita outcomes by interpolating quarterly populations from annual ACS 5-year estimates.

Both case-type distributions and the behavioral effect subset are less relevant in circuit courts, which deal overwhelmingly in felonies. Instead, I include a circuit-court specific measure for evidentiary effects: the number cases (per 100,000 residents) entering the court via appeal. In panel A of Table 2, I show the average values of case filings across Virginia courts. While shares of multi-charge cases are qualitatively similar for adopters and non-adopters, behavioral effect cases arise more frequently in treated courts.

To test for evidentiary and time use effects, I restrict the sample to only felonies and misdemeanors, and exclude probation violations, bond violations, and other charges that directly arise from prior interactions with the courts. From the remaining cases, I construct quarterly court-panels restricted by case type to account for the differing processes and potential outcomes across case types. This generates three panels: district misdemeanors, district felonies, and circuit felonies.

Key outcomes relate to case dispositions (guilty, prosecutor dropped charge), sentencing, and case duration. Panel B of Table 2, shows the share of cases reaching each of these dispositions varies between panels, but is similar across treatment groups within each panel. Felonies can resolve in the district court, or be "certified" to advance to the circuit court for further litigation. As such, I also test for changes in the share of district court felony cases that are certified to the circuit court. I use the share of cases in which the defendant is sentenced to a positive amount of time in jail or prison as the sentencing outcome. Finally, I construct the share of cases that received a disposition within 1 year of filing as a measure of case duration. A precise disposition date variable is available at the circuit but not district court levels, so for district court analyses I use the most recent hearing date instead.[10]

This disposition timing variable both measures case duration and fulfills a second role in the circuit court analyses; I selected the sample window to allow cases a full year to resolve, but some – especially more complex circuit court felonies – will take longer. Across the sample period, the average circuit court resolves over 80% percent of felonies within 1 year of filing. To mitigate the effects of the attrition of more complex cases toward the end of the sample period, I condition circuit court outcome variables on having been observed within 1 year of the filing date. The case length variable will alert me to compositional changes in my sample stemming from this timing criteria.[11]

The predicted effect of body-worn cameras on each of these outcomes is conceptually ambiguous: attorney time use effects are expected to worsen outcomes for defendants by forcing a reallocation of attorney time from higher to lower marginal value activities, which would correspond to an increase in both guilty and incarceration shares. However, it is unclear who benefits from evidentiary effects on average. If body-worn camera video typically supports defendant narratives, then the evidentiary channel may partially or fully offset the time use effects. This tension across effects is also present in the dropped charges outcome; prosecutors may be eager to speed cases along in plea negotiations by offering to drop charges, or may find their negotiation position strengthened by the new video evidence and do so less often. Attorney advocates predict dominant time use effects should lengthen the amount of time between case filings and resolutions.

[10]This will cause an overestimate of the time to case resolution, particularly for cases for which a defendant was sentenced to probation, but should serve as an effective proxy.

[11]In practice, this restriction is not influential, as I show in Table C.5

5 Methods

I test for the effects of body-worn cameras on policing and the courts using a difference-in-differences strategy based on the rollout of body-worn cameras across law enforcement agencies in Virginia. Law enforcement adopt body-worn cameras at different points in time, and heterogeneity in effects is plausible due to changes in salience at the policing stage and attorney adaptation within the courts. In light of this, I implement the imputation estimator developed by Borusyak et al. (2021) to estimate the following model:

$$Y_{lt} = \alpha + \tau D_{lt} + \delta X_{lt} + \gamma_t + \lambda_l + \epsilon_{lt} \tag{1}$$

Here, $D_{lt} = 1[t \geq T_l]$ is an indicator that takes the value of one for adopting localities (courts) during or after the quarter of adoption. The vectors γ_t and λ_l account for quarter and locality-specific fixed effects, while X_{lt} contains covariates that vary across locality (l) and quarter (t)– such as the unemployment rate and, for case process/resolution outcomes, additional controls for race/gender shares, charges per case, and number of cases in the court. The key parameter of interest is τ, the effect of body-worn camera adoption on the outcome of interest Y_{lt}.

The BJS imputation estimator operates by first estimating the fixed effects using only untreated observations, and then using these to construct counterfactual outcomes for treated observations. That is, using only untreated observations, I estimate model (2) to obtain $\hat{\lambda}_l$ and $\hat{\delta}_t$.[12]

$$Y(0)_{lt} = \lambda_l + X'_{lt}\delta_t + \epsilon_{lt} \tag{2}$$

Estimates $\hat{\lambda}_l$ and $\hat{\delta}_t$ are then substituted into model (1) to estimate each $\hat{Y}(0)_{lt}$ and calculate each $\hat{\tau}_{lt} = Y_{lt} - \hat{Y}(0)_{lt}$. I aggregate these in two ways: an overall ATT for the four quarters following adoption, comparable to the intended ATT from a constant effect two way fixed effects model, and dynamic event study estimates for each quarter individually. A limited subset of rarer outcomes (behavioral effect cases, appeals) are more appropriately modeled using a Poisson specification, due to non-trivial shares of quarters with few or 0 qualifying cases. For these, I reproduce the analogous imputation process for Poisson models instead of the above ordinary least squares. I cluster standard errors at the court-level, and produce analogous bootstrapped standard errors for the Poisson models.[13]

5.1 Identifying Assumption

This empirical strategy relies on a parallel trends assumption, for which I provide empirical support using the test proposed by Borusyak et al. (2021). This entails estimating an expanded version of model (2) that includes a vector W_{lt} of indicator variables for quarters leading up to adoption (model (3)), and then conducting a joint significance test of the leads.

[12] $X'_{lt}\delta$ nests the time fixed effects but also includes time-varying controls, such as defendant race and sex shares and the prevalence of different case types.

[13] Estimation was primarily conducted using the did_imputation command in Stata. LLMs were used as an assistive tool generate limited code for this project, and to improve formatting (i.e. of citations, tables, etc.).

$$Y(0)_{lt} = \lambda_l + X'_{lt}\delta_t + W'_{lt}\gamma_t + \tilde{\epsilon}_{lt} \tag{3}$$

This test sidesteps the issues raised in Roth (2022) regarding survivor bias for estimates that pass common pre-trends tests[14] Using a two-year pre-period and 10% significance level, all main results and nearly all heterogeneity results satisfied this pre-trends test. Event study plots in Appendix B further demonstrate that treatment and control group outcomes did not systematically diverge prior to treatment periods.

One possibility that this test cannot fully account for is that the mechanism for treatment assignment introduces an undetected violation of the parallel trends assumption. Body-worn camera adoption is not random by design or in practice: police departments choose to implement a body-worn camera program. Table 2 shows that the resulting set of treated courts are more likely to be in populous areas with more cases, and a higher share of Black defendants. To this point, I argue that body-worn camera adoption decisions are plausibly exogeneous to court processes and resolutions. The LEMAS-BWCS survey shows that law enforcement adopted the technology in pursuit of improvements to police interactions rather than desired changes in the courts. Fifty of the 53 adopting agencies surveyed in Virginia cited non-court motivations as the primary reasons for using the cameras. Within my samples, the baseline differences between treated and untreated courts shown in Table 2 are largely related to community characteristics rather than court processes or resolutions. The shares of cases with a guilty disposition, dropped, or amended charges are all similar between treated and untreated courts.

For police, a primary limitation to body-worn camera adoption during this time period was the financial costs to camera acquisition, training, and data retention. These costs create administrative hurdles even among eventual adopters, as shown in Kim (2025). I use this fact to supplement the primary analyses with one using only not-yet treated localities as controls. To do so, I truncate the sample period and use the 2018 adopters as "never treated" controls. This strategy – used by Kim (2025) in his national study of the effects of body-worn cameras on policing– allows for adoption itself to be endogeneous, while the timing of adoption should not be. I present these results in Appendix C.

6 Results

6.1 District Court

Panel A of Table 3 presents the results for district court case filing effects. In total, these results suggest that body-worn cameras induce behavioral effects at the policing stage: new filings of "behavioral effect" cases decline by approximately 5 cases per 100,000 residents, or nearly 10% of the analytic sample mean. Overall filing rates similarly show a negative point estimate, but are smaller in magnitude (approximately 2% of the mean), not statistically significant, and more sensitive to modeling choices (Table C.3). Consistent with moderating behavioral effects, cases become marginally less complex; I estimate a 0.6 percentage point

[14] See Wooldridge (2021) for equivalence between this method and a common pre-trends test in a fully saturated model.

decline in the share of multi-charge cases, which is equivalent to approximately 3% of the mean. Relative to felonies, the share of misdemeanors declines slightly – suggesting that insofar as behavioral effects reduce case filings, they are more likely to cull lower-level criminal charges from the courts than to reduce the severity of high-level offenses.

Panel B of Table 3 shows results for case process and resolution analyses. Although case filings marginally changed as a result of body-worn cameras, I do not find evidence that this translated to economically or statistically significant effects on case outcomes, conditional on filing. Body-worn camera introduction does not affect the probability a case resolves with a guilty disposition, or the related probability that the defendant is sentenced to incarceration. For felonies, these point estimates each amount to approximately 1% of the mean. The estimates are precise enough to rule out moderate changes in these outcomes. For misdemeanors, for which guilty dispositions carry a lower incarceration probability, I estimate an insignificant decline in guilty dispositions and incarceration of 0.6pp (0.8%, 3% respectively). These cases are more common than felonies, and 95% confidence intervals exclude decreases larger than 2.4% for guilty dispositions. Importantly, I can also rule out economically meaningful increases in these resolution outcomes: 95% confidence intervals exclude incarceration increases greater 2% and guilty disposition increases greater than than 0.8%.

The processes by which these outcomes are realized also remain stable after body-worn camera adoption: I do not find evidence that prosecutors drop or amend charges more often for either felonies or misdemeanors. Felony cases are no less likely to be certified to the circuit court (0.1% decline; insignificant). Indicators for whether a case resolved in either six months or one year show signs congruent with slowdown– all are negative – but are not statistically significant and reflect economically small magnitudes ranging from 0.3 to 1%. Together, this suggests that body-worn cameras did not meaningfully slow down court processing times.

In the context of the case filing results, the totality of the district court findings suggest that body-worn camera effects are concentrated in the pre-adjudication stages, driven by behavioral effects rather than evidentiary or attorney time use effects. The strongest effects I find lie in a narrow subset of filings ("behavioral effects" cases) that are unlikely to measurably change subsequent court processes alone.

6.2 Circuit Court

By the time a case advances to the circuit courts both police and court actors have interacted with it. Although the circuit court has jurisdiction over felonies, preliminary hearings for these cases take place at the district court level and so evidence, negotiations, and pleas routinely result in charges being dropped, dismissed, or otherwise disposed before reaching the circuit courts. Because of this, circuit court filings are a selected set of cases that persisted past district court-level off-ramps; changes to these filings can reflect a combination of all three prospective channels for body-worn camera effects. Yet while the mechanisms for any changes are difficult to disentangle at this level, the aggregate changes nonetheless describe policy-relevant effects. Felony convictions carry sentences of at least a year of imprisonment, restrict post-release labor market opportunities, and – until recently – caused permanent disenfranchisement of convicted offenders in Virginia (Brennan Center for Justice, 2018).

I begin with a similar set of case filing outcomes at the circuit court level as at the district, including measures for the overall number of new cases and the share of cases that include multiple charges. Behavioral effect cases rarely advance to the circuit court, so I omit this outcome. Instead, I evaluate an outcome unique to the circuit court that reflects carry-through evidentiary and time use effects from the district court: the frequency of appeals. Panel A of Table 4 presents these results. Similar to the district court findings, point estimates for effects on case filings are negative, small (approx. 2.5% of the mean), and not statistically significant. They allow me to rule out moderate increases of about 5% and declines exceeding 10% at 95% confidence. Any declines are not the result of changing rates of appeals which I estimate to be similarly small (1.4% of the mean), statistically insignificant, and positive. Conditional on filing, body-worn camera adoption decreases the probability a case has multiple charges by 1.6pp (3.5%).

Case process and resolution results confirm the district court finding of no effects among the subset of filed cases. Panel B of Table 4 shows no effect of body-worn camera adoption on the share of cases with dropped (0.3pp or 1%) or amended (0.4pp or 3%) charges. The estimates are precise enough to rule out moderate but not small effects. For key case resolution outcomes, the estimated effects are similarly small: the share of cases with a guilty disposition declines by an insignificant 0.7%, and incarceration shares decline by an insignificant 1.1%. For these outcomes, 95% confidence intervals exclude declines larger than 3.4 and 4.1%, respectively, or increases larger than approximately 2%. Similar to the district court analog, I also do not find significant evidence of court slowdowns.

6.3 Temporal dynamics

I show the temporal decompositions of the estimated effects in a series of event study plots in Appendix B. Each plot shows the quarterly treatment effects for each subsample for one year after treatment as well as the eight pre-treatment periods from the pre-trends test described in section 5. In district courts, behavioral effect cases show an immediate decline in filings in quarter 0, the quarter of adoption. This decline reflects a level shift in the outcome at the time of adoption that persists throughout the following year. In contrast, plots for overall case filings and compositional changes support a practicably null effect of body-worn cameras on the broader caseload. Measured changes are small in magnitude, and overall filings do not show a persistent downward trend. For process and resolution outcomes, with the exception of the 6 month-disposition variable, the post-periods show no discernible trend: the estimated null results do not appear to mask any meaningful temporal dynamics. The slight slowdown in time-to-disposition for district court felonies is an exception to this pattern; while the point estimate is small and insignificant, it appears to reflect a sustained decrease in the share of cases disposed within 6 months of filing. Circuit court process and resolution outcomes again mirror the district court null result, but filing plots provide suggestive evidence of a slight sustained decline in multi-share and total case filings.

The immediacy of the behavioral effect is consistent with rapid salience of body-worn cameras during police interactions. On the other hand, there could be delays in effects for court processes and resolutions if court actors learn to use the new data over time. Such dynamics could result result in short-term one-year effects overstating the behavioral effects of cameras, if officers/the public become desensitized to cameras, or understate the adjudicatory effects if effects develop over time. For this reason, I provide extended horizon estimates in Table D.1 and Table D.2. For this analysis, I restrict the sample to untreated and treated courts with at least two years of post-adoption case data to maintain a temporally balanced panel. Overall, point estimates show quite similar effects: the behavioral effect case estimate in the extended sample is qualitatively slightly larger at -6.6 cases per 100,000 compared to -5.1 in the one-year specification. The magnitude of the estimate for the measured decrease in misdemeanors remains identical, at 0.5*pp*. I continue to find null effects for all case process and resolution outcomes at the district court level, although the point estimate for the guilty disposition outcome for district court felonies is larger – growing from 1% to 3.7% of the mean. Circuit court outcomes similarly show nominally larger point estimates in the extended horizon, which continue to translate to statistically insignificant and economically small effects.

6.4 Robustness and Alternative Estimands

These results are broadly robust to a range of modeling choices, which I present in C. As described above, overall district court case filings show some sensitivity to estimation method – in particular, the specification using only adopters shows a reversal in sign, but remains small and statistically insignificant. Other outcomes show stable results across specifications, at times with varying precision. Specifically, results are similar when I define treatment status using a 50% treatment threshold, (Table C.1; Table C.2). Results are also similar when applying alternative estimators, restricting the control group to only adopters, varying the sets of controls, and including later treated units that were previously omitted to preserve a balanced panel structure (Table C.3; C.4). In Table C.5, I show that the circuit court results are not sensitive to the restriction that cases resolve within 1 year of filing.

I also present results for two alternative estimands. My primary estimand pertains to the estimated effect of body-worn cameras on an adopting court, constructed using equally weighted court-level outcomes. This is a policy-relevant unit of observation that aligns with the level at which I observe treatment; I evaluate the system-level effects of body-worn cameras on court outcomes, and observe which courts – not specific cases– are exposed to treatment. Results are intended to inform of the expected effects of court-level adoptions. Even so, population-weighted estimands – which pertain to the overall effects of body-worn cameras on Virginia in the study period– are relevant for some uses. I show in Table D.3 that district court estimates are similar when weighting treatment effects by baseline population. For circuit courts, shown in Table D.4, process and resolution outcomes are similarly stable, but I observe a decline in the number of filings and appeals. In Figure D.1 and Figure D.2, I graphically show the court-level decomposition for these two outcomes.

Secondly, my primary results present case process and resolution outcomes *conditional* on a case entering the courts – that is, conditional on there being an opportunity for time use and evidentiary effects to take place. An alternative estimand of interest could be the population-level rates of case outcomes, such as guilty dispositions. In Table D.5 I supplement these with unconditional per-capita analogs, defining outcomes per 100,000 residents in the court's geographic jurisdiction. This alternative specification produces similar results for district court felonies, showing no effects of body-worn cameras on any outcomes. For misdemeanors, however, incarceration declines by approximately 8% from baseline. In circuit courts, directionally, effects in the main conditional specification and the unconditional supplement are congruent. However, the unconditional magnitudes are more pronounced. This is consistent with the multi-stage filtering that precedes cases entering this court. Panel B of Table D.5 shows a statistically significant, mild decline in both guilty dispositions and incarceration (approx. 6% from the mean), partly driven by a similarly sized slowdown in resolutions.

7 Key Heterogeneity Analyses

The overarching finding – calculated across multiple case types and courts – that body-worn cameras have limited effect on case filings and resolutions is both robust and surprising: practitioners emphatically report that inputs to the criminal justice system changed when police began using body-worn cameras. An influx of camera footage added data to the courts and an additional job responsibility for attorneys. Virginia created a committee to document these court changes, and even introduced legislation limiting the number of cameras per prosecutor. And yet, I find no aggregate effects of body-worn cameras on most court outcomes. This disconnect raises additional questions about how these null results came to be.

One explanation is that body-worn cameras are simply a smaller shock to the system than practitioners and advocates perceive them to be: case processes and resolutions could be sticky and unresponsive to changes in attorney time use or the magnitude of the noise reduction produced by body-worn cameras. In this scenario, body-worn cameras could be strongly influential only in a small subset of cases that do not change aggregate processes and resolutions – bridging the sentiment of practitioners with the results above. An alternative explanation for the null results is that various channels of body-worn camera effects offset one another.

In this section I explore heterogeneity in effects for two key subsets of the data to evaluate these null results. First, I use variations in the likelihood an offense is captured on body-worn camera video to test the offsetting effects hypothesis. Then, I test for heterogeneity in effects based on a key defendant demographic characteristic: race. Body-worn camera adoption in the U.S. is tightly linked to broader concerns with racial disparities in policing. I test whether the subset of Black defendants experienced differential changes to case filings, processes, or resolutions due to body-worn camera adoption.

7.1 Revisiting Evidentiary and Time Use Channels

In the primary results, case filing outcomes suggest that body-worn camera behavioral effects do not induce widespread changes in caseloads or characteristics. The subsequent case process and resolution outcomes also show no effect, this time from the combined influence of evidentiary and attorney time use channels.[15] One driver of the null result could be that

[15]If the cases entering the courts appreciably change after body-worn camera adoption due to policing changes, then estimates of body-worn camera effects on case outcomes may reflect not only the court-based evidentiary and time use effects but also selection effects from policing-based case changes. However, in

these two channels offset one another: evidentiary benefits cancel attorney time costs in the aggregate. I test this counteracting effects hypothesis using variation in the likelihood that footage is available for a case. While I cannot observe whether body-worn camera video was available for or used in a specific case, I can identify a subset of charges more likely to take place in front of an officer. These cases should be more likely to have relevant body-worn camera video, and thus experience both evidentiary and time use effects. However, this does not mean that these are the only cases affected by body-worn cameras. Attorneys often work multiple cases at a time. If attorneys substitute their hours across cases to the case activities with the highest marginal benefit then at times they will substitute their work hours from cases without body-worn camera video toward a case with footage.[16] If this happens systematically, then a divergence in case outcomes would suggest a dominant evidentiary effect for cases with video.

To assess this divergence, I partition cases based on the underlying charges. I identify cases as "BWC-type" if they include any charge likely to take place in view of an officer. These include the behavioral effect charges from prior analyses, but also DUI/DWI, concealed weapons, and possession of weapons or drugs. An individual receives a concealed weapon or possession charge because an item was observed in the presence of an officer, and thus if the officer is wearing a camera it is likely there will be footage associated with the charge. Such charges differ from the subset of "behavioral effect charges" because the alleged offense can be initiated prior to interaction with an officer. For example, one can forgo an offense of assault on an officer by refraining from making physical contact, but cannot choose to no longer be in possession of an illicit substance after an interaction with an officer has begun. I include DUI/DWI based off of the input of numerous law enforcement officers throughout Virginia who independently volunteered this as an example of a charge likely to be affected by body-worn camera footage. Under this partitioning, I observe an average of 36.7 BWC-type and 47.4 non-BWC-type cases per quarter in circuit courts; 45.9 BWC-type and 56.7 non-BWC-type felony cases in district courts; and 115.4 BWC-type and 685.3 non-BWC-type misdemeanor cases in district courts.

My combined analytic data set for this analysis then consists of two observations per court-quarter: one for each case-type group. Applying the imputation estimator to this pooled data set, I estimate the group-specific treatment effects for all primary outcomes and present the differences in these effects in Table 5. Because the partitioning will result in an increase in court-quarters with no cases, I also present bounds on the group-specific treatment effects in Figure C.1 under alternative scenarios for the missing data. While some differenced outcomes are noisy, leading to large error bands around point estimates, overall I do not find compelling evidence for a divergence in court processes or resolutions. I find some evidence for a small differential reduction in guilty dispositions (approximately 3% of the BWC-type mean), but interpret this result cautiously because it is an outlier among the

practice, the cases show limited observable changes, and both conditional and unconditional case process and resolution results are similar – suggesting limited selection effects.

[16] The attorney optimizes time use in a way that incorporates both defendant interests and her own. Some activities that yield lower returns to defendant outcomes may nonetheless have higher returns in this framework. For example, even if the defender is convinced that, irrespective of effort, a defendant will be convicted and sentenced harshly she must nonetheless complete certain tasks in order adhere to professional standards.

totality of the results. It is unlikely that the counteracting effects hypothesis holds in the aggregate; body-worn camera footage appears to have minimal effects on the case processes and resolutions as a whole.

7.2 Heterogeneity by Defendant Race

Public advocacy for body-worn cameras grew against the backdrop of a police shooting in Ferguson, Missouri that ignited large-scale protests centered on racial disparities in policing (BBC, 2014). Black adults in the U.S. persistently express less confidence in the police than do white adults, and polls show that this gap grew throughout the 2010s (AP-NORC, 2015, Jones, 2020). Beyond perception, numerous studies document racial disparities and discrimination in the criminal justice system, including in policing (Antonovics and Knight, 2009, Horrace and Rohlin, 2016), pretrial release (Arnold et al., 2018), convictions and jury deliberations (Abrams et al., 2012, Anwar et al., 2012, Bjerk and Helland, 2020, Flanagan, 2018), and sentencing (Alesina and La Ferrara, 2014). Additional research shows that some policies intended to ameliorate these disparities and their effects can unintentionally exacerbate them (Doleac and Hansen, 2020).

Because body-worn cameras are intended by many to especially improve police interactions for Black individuals, I test for differential effects of body-worn cameras on Black defendants. I employ the same techniques used in the previous section of this paper to test for differential effects, with cases partitioned into by the reported race of the defendant. Specifically, I test for differences in case processes and resolutions for Black and non-Black defendants.[17]

First, I note that the mean differences in outcomes across the two racial groups are small within this sample. For most outcomes, the difference between average Black and non-Black defendant outcomes across courts is less than one percentage point.[18] The largest observed difference across outcome domains is in the share of defendants sentenced to serve time for misdemeanors; Black defendants are sentenced to time 3.8 percentage points more often than non-Black defendants. Yet, Table 6 shows no differential effect in this outcome for Black defendants. Where a significant negative point estimate would suggest body-worn cameras closed the sentencing gap, I instead find a statistically insignificant but positive point estimate. Overall, across the range of outcomes and court levels, I do not find evidence that body-worn camera adoption differentially improved case outcomes for Black defendants.

[17] A challenge for this analysis stems from the racial homogeneity within many rural localities. For example, at the circuit court level, Black defendants make up 45% of all cases while white defendants comprise 53%. However, less populous localities routinely show in excess of 90% non-Black defendants. In Figure C.2, I present alternative bounds on the differenced effects when imputing values – such as 0, 1, or the locality-group mean – for observations with no qualifying cases. Overall, these suggest small or null differential effects under plausible imputations.

[18] This parity should be understood in the context of case filing differences. At the district court level, these show an over-representation of Black defendants relative to population share of approximately 8 percentage points. Of note, a test for the effect of body-worn cameras on the share of these filings with a Black defendant is small and statistically indistinguishable from 0 (0.1pp; 95% confidence interval [-0.3, 0.6]).

8 Conclusion

Body-worn cameras have become a key tool in a public push for transparency and accountability for police officers. However, while law enforcement agencies equipped officers with this recording technology, attorneys and other court actors grew concerned about unintended consequences of the data influx from body-worn cameras. The results of this study may ameliorate these concerns. Using a rich data set containing detailed charge-level information for criminal charges filed in Virginia courts between 2009 and 2019, I find body-worn cameras have an overall limited effect on new criminal cases. While an important subset of charges become less prevalent after police begin using body-worn cameras, indicating a role for behavioral effects before cases enter the courts, overall caseloads show limited changes in quantity or composition.

Conditional on filing, I find that body-worn camera adoption does not adversely affect criminal defendants. Defendants are found guilty and sentenced to incarceration at similar rates after police start to use body-worn cameras. This finding cannot be attributed to compositional changes in cases stemming from changing case characteristics, and is robust to the inclusion of various charge characteristic controls. This result is surprising: body-worn cameras generate hours of evidence and attorneys in Virginia report that they view this footage at substantial time costs. However, neither the evidentiary value of the footage nor the reallocation of attorney time to review it appear to affect criminal cases on average. While concerns over racial disparities in policing have been an integral part of the public discourse around body-worn camera adoption, I also do not find convincing evidence of differential policing or court effects for Black defendants.

Overall my results suggest that body-worn camera effects on policing and the courts are concentrated in rare, but potentially high-value interactions. Existing research shows that benefits such as reduced use-of-force in these interactions exceed the costs of obtaining and maintaining cameras (Williams Jr. et al., 2021) for police. Combining this prior result with my own findings that body-worn cameras do not substantially alter outcomes in the courts, it appears that expanded body-worn camera adoption will produce net benefits to the criminal justice system with both the costs and benefits accruing primarily to law enforcement and law enforcement interactions.

Although I use rich criminal case data, there are two considerations necessary to place this paper in its proper context. Just as I find body-worn cameras affect only a small subset of police interactions, it is possible that body-worn cameras are deeply influential in a small subset of criminal cases not captured in aggregated analyses. Additionally, even the richest criminal case data provide merely a snapshot of the broader costs and benefits borne by actors in the criminal legal system. Additional outcomes such as perceptions of fairness in the justice system would also provide valuable insight into the holistic effects of body-worn cameras. Such outcomes can be influential; the expressed attorney concerns described in the background of this paper eventually culminated in legislative efforts to increase funding for Commonwealth's Attorney (prosecutor) offices in Virginia. Where implemented, this would exacerbate funding differentials between indigent defenders and prosecutors in the years to come. In this context, the counter-intuitive finding of limited systemic effects suggests that such resource allocations are not necessary to balance body-worn camera effects. However, the margins at which additional workload or resourcing can affect measurable change in criminal courts remains a critical question for future study.

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9 Tables and Figures

Table 1: Case Roadmap

Stage	Case Activity	Entities Involved	Activity Summary, *Body-worn camera role*
1	(Alleged) Offense	Defendant Police	Defendant allegedly commits offense; may or may not be seen by police. *Video may be captured*
2	Summons issued/ Arrest and booking	Defendant Police	Officer may release defendant with summons for later court date or arrest defendant. *Video captured*
3	Warrant/ Charge issued	Defendant Police Magistrate	If arrested, defendant appears before a judicial officer to determine whether charges proceed. *None*
4	Court hearings, preparation	Defendant Judge Attorneys	Court hears evidence, disposes charges, pronounces sentences. Attorneys negotiate plea agreements, argue for preferred dispositions and sentences. *Video reviewable*

Table 2: Comparison of treated and untreated localities at early-sample baseline

	District		Circuit	
Panel A: Case Filings	Untreated	Treated	Untreated	Treated
Cases	7433.83	8710.94	178.43	254.33
(per 100k)	(318.66)	(554.59)	(8.59)	(8.10)
Behav. Effect Cases	41.62	67.69	8.13	11.52
(per 100k)	(1.99)	(2.45)	(4.68)	(4.25)
Has Multiple Charges	0.17	0.18	0.42	0.38
	(0.01)	(0.00)	(0.02)	(0.01)
Num. Felony/Misd.	1061.05	1315.42	60.54	131.90
	(214.14)	(96.35)	(6.01)	(9.90)
Appeals	–	–	59.74	102.24
(per 100k)	–	–	(2.53)	(6.74)
Share Black	0.18	0.27	0.24	0.40
	(0.01)	(0.01)	(0.02)	(0.01)
Share Female	0.34	0.36	0.24	0.25
	(0.00)	(0.00)	(0.01)	(0.00)
Local Pop.	69592	64459	40129	66023
	(15015)	(5089)	(4267)	(5199)
N	132	300	112	292

Panel B:	District Misd.		District Fel.		Circuit Fel.	
Case Processes & Outcomes	Untreated	Treated	Untreated	Treated	Untreated	Treated
Amended	0.18	0.18	0.16	0.16	0.14	0.13
	(0.01)	(0.00)	(0.01)	(0.01)	(0.01)	(0.00)
Sentence Time	18.67	23.61	65.77	80.05	2647.90	2385.59
	(0.89)	(0.83)	(3.88)	(2.91)	(186.05)	(84.42)
Sentenced to Time	0.17	0.21	0.27	0.28	0.71	0.73
	(0.01)	(0.00)	(0.01)	(0.01)	(0.01)	(0.01)
Fined	0.67	0.67	0.12	0.13	0.10	0.11
	(0.01)	(0.01)	(0.01)	(0.00)	(0.01)	(0.01)
Dismissed	0.19	0.17	0.10	0.10	0.06	0.04
	(0.01)	(0.01)	(0.01)	(0.00)	(0.01)	(0.00)
Charge Dropped	0.09	0.09	0.37	0.39	0.23	0.25
	(0.01)	(0.00)	(0.01)	(0.01)	(0.02)	(0.01)
Guilty	0.74	0.75	0.33	0.32	0.74	0.74
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Has Multiple Charges	0.32	0.32	0.58	0.56	0.52	0.45
	(0.01)	(0.00)	(0.01)	(0.01)	(0.02)	(0.01)
Num. Charges	1.46	1.46	2.41	2.33	2.74	2.28
	(0.01)	(0.01)	(0.05)	(0.03)	(0.14)	(0.04)
Num. Cases	968.61	1090.04	92.83	129.21	53.78	102.52
	(196.42)	(75.10)	(18.28)	(10.31)	(4.45)	(6.88)
N	132	296	132	296	112	292

Note: Quarterly-court averages based on 2009 case characteristics from locality-level panel. The treated group are localities that adopted by Q2, 2018. Share variables for counties with no charges filed in a given quarter are coded as missing.

Table 3: District Court: Case filings, processes, resolutions

Panel A: Filings	Cases Filed (per 100k)	Behav Effect (per 100k)	Multicharge	Share Misd.	Share Infract.
Treatment Effect	-144.188	-5.152*	-0.006*	-0.005**	0.005
District	(312.315)	(2.884)	(0.004)	(0.002)	(0.005)
Mean	7451.271	53.380	0.185	0.929	0.711
N	3353	3353	3353	3353	3353

Panel B: Case processes and resolutions	Prosecutor Dropped	Case Certified	Guilty	Sentenced to Time	Amend	Disposition 1 Year	Disposition 6 Mos
District Felonies							
Treatment Effect	-0.001	-0.001	0.003	-0.002	0.001	-0.006	-0.010
	(0.010)	(0.013)	(0.012)	(0.011)	(0.008)	(0.005)	(0.009)
Mean	0.391	0.571	0.302	0.250	0.150	0.966	0.880
N	3306	3306	3306	3306	3306	3306	3306
District Misdemeanors							
Treatment Effect	0.003	–	-0.006	-0.006	0.006	-0.003	-0.005
	(0.003)	–	(0.006)	(0.005)	(0.005)	(0.003)	(0.005)
Mean	0.093	–	0.750	0.193	0.197	0.957	0.890
N	3353	–	3353	3353	3353	3353	3353

Results from BJS imputation estimator. Cluster-robust standard errors in parentheses. Behavioral effect estimates analogously produced using poisson regression and bootstrapped standard errors. Share of misdemeanors is presented relative to sum of misd. and felony cases. Panel A controls for local unemployment rate. Panel B Controls include average number of charges per case, shares of female and Black defendants, count of cases in the court, and the locality unemployment rate. * p<0.10, ** p<0.05, *** p<0.01. † indicates pre-trend test p<0.10.

Table 4: Circuit Court: Case filings, processes, resolutions

Panel A: Filings					
		Cases Filed (per 100k))	Appeals (per 100k)	Multicharge	
Treatment Effect		-5.852	1.204	-0.016**	
		(8.832)	(4.082)	(0.008)	
Mean		238.122	85.756	0.463	
N		3080	3080	3073	
Panel B: Case processes and resolutions					
	Prosecutor Dropped	Guilty	Sentenced to Time	Amend	Disposition 1 Year
Treatment Effect	-0.003	-0.005	-0.008	-0.004	-0.003
	(0.011)	(0.010)	(0.011)	(0.007)	(0.009)
Mean	0.261	0.728	0.721	0.132	0.821
N	3062	3062	3062	3062	3062

Results from BJS imputation estimator. Cluster-robust standard errors in parentheses. Behavioral effect estimates analogously produced using poisson regression and bootstrapped standard errors. Panel A controls for local unemployment rate. Panel B Controls include average number of charges per case, shares of female and Black defendants, count of cases in the court, and the locality unemployment rate. * p<0.10, ** p<0.05, *** p<0.01. † indicates pre-trend test p<0.10.

Table 5: Heterogeneity by BWC-type Case

	Prosecutor Dropped	Case Certified	Guilty	Sentenced to Time	Amend	Disposition 1 Year	Disposition 6 Mos
Panel A: District Felony							
Diff. Effect	0.023†	-0.013	0.015	0.005	0.004	0.002	-0.016
	(0.023)	(0.024)	(0.022)	(0.020)	(0.017)	(0.008)	(0.017)
Mean (BWC^1)	0.465	0.644	0.349	0.278	0.132	0.960	0.852
Mean (BWC^0)	0.343	0.514	0.274	0.237	0.165	0.971	0.906
N	6545	6545	6545	6545	6545	6545	6545
Panel B: District Misdemeanor							
Diff. Effect	0.011†	–	-0.025*	-0.010	-0.005	0.011	0.002
	(0.011)	–	(0.014)	(0.014)	(0.015)	(0.009)	(0.013)
Mean (BWC^1)	0.231	–	0.788	0.576	0.136	0.884	0.763
Mean (BWC^0)	0.071	–	0.742	0.132	0.205	0.968	0.911
N	6703	–	6703	6703	6703	6703	6703
Panel C: Circuit							
Diff. Effect	0.030	–	-0.015	-0.018	0.004	0.003	
	(0.023)	–	(0.018)	(0.019)	(0.015)	(0.018)	
Mean (BWC^1)	0.265	–	0.702	0.695	0.089	0.775	
Mean (BWC^0)	0.253	–	0.751	0.743	0.165	0.860	
N	6050		6050	6050	6050	6050	

Results from BJS imputation estimator. Cluster-robust standard errors in parentheses. Controls include average number of charges per case, court gender and race shares, count of cases in the court, and the locality unemployment rate. BWC^1 cases are those more likely to have relevant video, and BWC^0 those less likely. * p<0.10, ** p<0.05, *** p<0.01. † indicates pre-trend test p<0.10. Difference rows report lincom test of ATT(BWC^1) − ATT(BWC^0). Means are reported separately for each subgroup; N is the full estimation sample.

Table 6: Case Outcome Heterogeneity by Race (Black − Non-Black)

	Prosecutor Dropped	Case Certified	Guilty	Sentenced to Time	Amend	Disposition 1 Year	Disposition 6 Mos
Panel A: District Felony							
Diff. Effect	0.039*	-0.007	-0.014†	-0.015	-0.018	0.003	0.019
	(0.020)	(0.025)	(0.021)	(0.020)	(0.017)	(0.011)	(0.017)
Mean (Black)	0.391	0.572	0.297	0.253	0.145	0.968	0.882
Mean (Non-Black)	0.390	0.566	0.306	0.252	0.158	0.965	0.880
N	6334	6334	6334	6334	6334	6334	6334
Panel B: District Misdemeanor							
Diff. Effect	0.001	–	-0.005	0.012	0.002	-0.001	-0.007
	(0.006)	–	(0.010)	(0.010)	(0.009)	(0.005)	(0.008)
Mean (Black)	0.092	–	0.766	0.217	0.190	0.960	0.894
Mean (Non-Black)	0.092	–	0.745	0.179	0.202	0.957	0.892
N	6650	–	6650	6650	6650	6650	6650
Panel C: Circuit							
Diff. Effect	-0.020	–	0.004	0.006	-0.001	0.001	
	(0.029)	–	(0.026)	(0.025)	(0.018)	(0.021)	
Mean (Black)	0.261	–	0.739	0.731	0.129	0.839	
Mean (Non-Black)	0.255	–	0.725	0.717	0.133	0.812	
N	5882	–	5882	5882	5882	5882	

Results from BJS imputation estimator. Cluster-robust standard errors in parentheses. Controls include average number of charges per case, court gender share, count of cases in the court, and the locality unemployment rate. * $p<0.10$, ** $p<0.05$, *** $p<0.01$. † indicates pre-trend test $p<0.10$. Difference rows report lincom test of ATT(Black) − ATT(Non-Black).Means are reported separately for each subgroup; N is the full estimation sample.

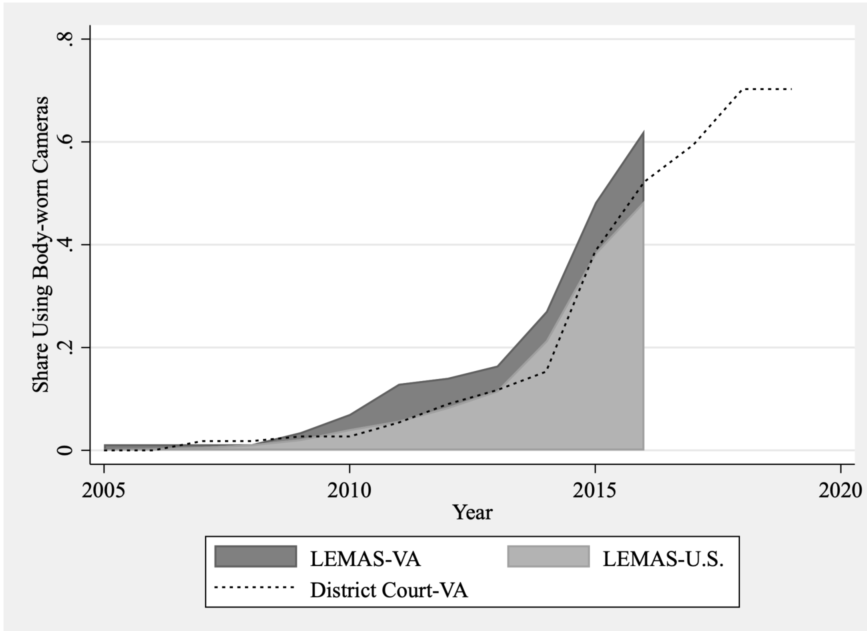


Figure 1: BWC adoption in the U.S. and Virginia. LEMAS-U.S. and LEMAS-VA shares reflect unweighted responses from a national sample of law enforcement agencies. District Court-VA shares are aggregated at the district-court level from major law enforcement agencies. Missing responses are omitted from the share calculations.

For Online Publication

A Data

In this appendix I further describe the data sources used in this paper as well as the decision rules I applied in preparing the data for analysis.

A.1 BWC Data

I constructed a body-worn camera adoption dataset using multiple sources of information on the timing of body-worn camera adoption by local law enforcement throughout Virginia. The three primary sources I used to construct this dataset were, 1) FOIA requests 2) Local news and agency websites 3) Non-FOIA personal contact with departments. Data collected through personal contact were typically collected in the exploratory stages of the project. Within the FOIA requests I asked for separate information for pilot programs, if applicable. Departments commonly use a testing or pilot phase in which a limited number of officers are given body-worn camera to use for a short time period to provide feedback to a department considering or planning to adopt body-worn camera on a larger scale. For example, one large department of over 200 officers piloted the technology with eight officers who had temporary use of the cameras. Other departments do not formalize this as a "pilot program" but begin by outfitting very few officers with cameras before establishing a department program. I do not treat these pilot and preliminary programs as adoptions.

From these sources I obtained information about body-worn camera implementation for 166 agencies throughout Virginia, reflecting complete body-worn camera adoption data for the major law enforcement agencies in 111 district court jurisdictions including 78 that adopted body-worn cameras by 2019 and 106 circuit court jurisdictions, 76 of which adopted by 2019. These comprise nearly 90 percent of state district and circuit courts in Virginia. I include a map of adopting jurisdictions is available. In the remaining 10 percent of localities, at least one major agency either failed to respond to the FOIA request or had incomplete records.

This set of 166 agencies is not exhaustive: there are hundreds of local law enforcement agencies throughout Virginia. Oftentimes multiple agencies operate within a single court jurisdiction. Because these agencies typically vary in force size and the size of the populations they serve, their individual influence on local courts also varies. For example, according to the 2008 Census of State and Local Law Enforcement Agencies, 15 Virginia departments had only one full time sworn officer while 35 departments had over 100. As such I defined a court jurisdiction to be treated when the first "major" local law enforcement agency operating in the court jurisdiction began using body-worn cameras, excluding small scale pilot adoptions.

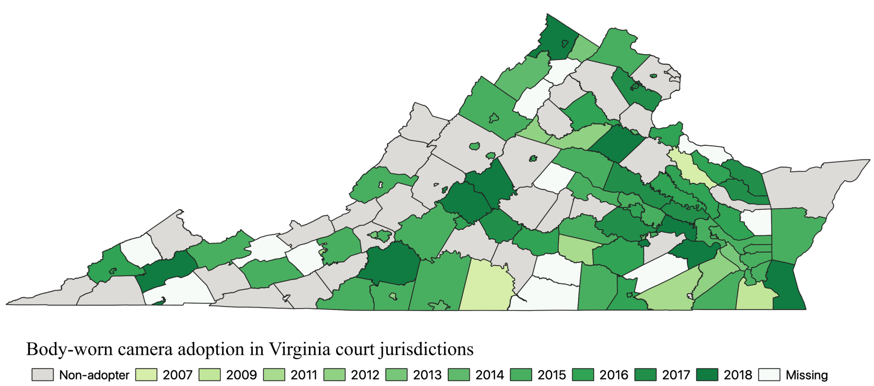
My primary specification used throughout the paper considers a law enforcement agency "major" within its locality if it is a policing organization that has jurisdiction over at least 25% of the locality's population *or* employs at least 25% of the locality's full time sworn officers amongst agencies with policing mandates. I also present appendix results with an alternative, 50% threshold. I used two sources of information to determine which agencies would meet these criteria:

Law Enforcement Force Size and Characteristics: I use policing role indicators and force size measures from the 2016 Law Enforcement Agency Roster (LEAR). The LEAR itself includes variables pulled from other sources. Thus the LEAR 2016 officer counts I use are counts from the 2008 Census of State and Local Law Enforcement Agencies (CSLLEA 2008). The population served by an agency is pulled from the 2014 UCR Population as listed in the FBI Police Employee Data from the same year.

The LEAR variable indicating policing activities is not always fully reflective of the mandate of an agency. Particularly in large and medium sized cities, it is common for both a police and sheriff's department to operate within city limits. However, the sheriff's department may be tasked with court security, civil processes, and jail security in contrast to the police department which engages in patrol and investigations. In many of these cases the LEAR population variable is missing, and the officer count may be substantially greater than the true number of officers engaging in policing activities. I omitted such agencies.

Locality Population Size: I use intercensal population estimates from the Weldon Cooper Center for Public Service, Demographics Research Group for estimates of the 2014 locality level population size. I developed crosswalks matching the counties and cities in these population estimates to the courts with jurisdiction over them. One city in my sample is split across two circuit court jurisdictions. In this case I applied half of the estimated population of the city to each relevant court jurisdiction.

I used intercensal population estimates rather than a sum of LEAR population estimates to head off potential issues with double counting in shared jurisdictions as well as missing data issues which could respectively inflate and deflate the denominators of the calculated shares. However, as a data check I compared the population shares calculated using a sum of LEAR populations to my primary share measure (using intercensal estimates). The departments classified as "major" were unchanged.



A.2 Case Data

I obtained charge data from VirginiaCourtData.com. To provide more clarity on the form this data takes, I include here an example of the web-based case information that the owner of this repository scrapes (Schoenfeld, 2017). To maintain the privacy of the defendant, I redacted information that could be used to easily identify this specific record online.

Case Number: CR08[redacted]	**Filed:** [redacted]	**Commenced by:** Reinstatement	**Locality:** COMMONWEALTH OF VA
Defendant: G[redacted]	**Sex:** Male	**Race:** Black (Non-Hispanic)	**DOB:** 10/01/****
Address: ARLINGTON, VA 22207			
Charge: VIOL PROBATION ON FEL OFF	**Code Section:** 19.2-306	**Charge Type:** Felony	**Class:** U
Offense Date: [redacted]	**Arrest Date:** [redacted]		

Final Disposition

Disposition Code: Sentence/Probation Revoked	**Disposition Date:** [redacted]	**Concluded By:** Revoked
Amended Charge:	**Amended Code Section:**	**Amended Charge Type:**

Jail/Penitentiary: Penitentiary	**Concurrent/Consecutive:** Sentence Is Run Consecutively With Another	**Life/Death:**
Sentence Time: 3 Year(s)	**Sentence Suspended:**	**Operator License Suspension Time:**
Fine Amount:	**Costs:** $658.00	**Fines/Cost Paid:**
Program Type:	**Probation Type:**	**Probation Time:**
Probation Starts:	**Court/DMV Surrender:**	**Driver Improvement Clinic:**
Driving Restrictions:	**Restriction Effective Date:**	
VA Alcohol Safety Action:	**Restitution Paid:**	**Restitution Amount:**
Military:	**Traffic Fatality:**	

Appealed Date:

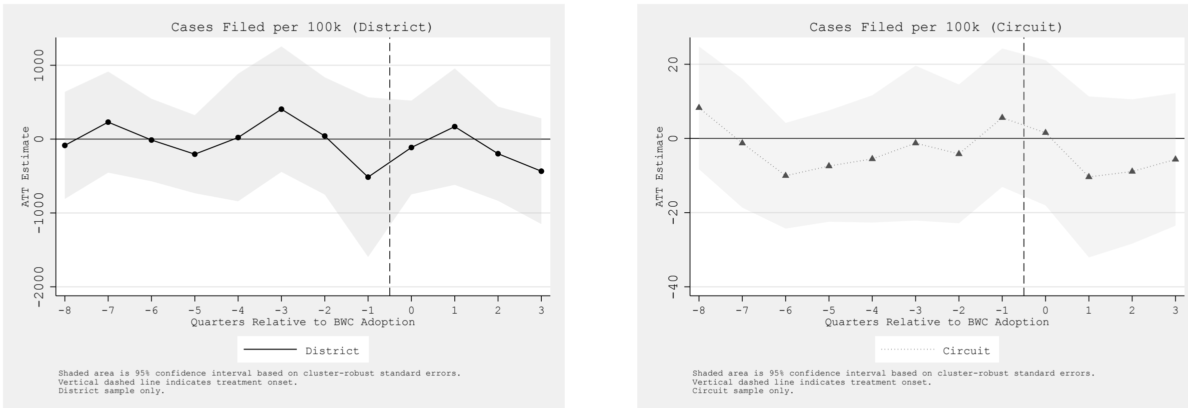
I aggregated this charge-level data to a case level before forming the court-level panel. To identify which charges were associated with a common case, I began by using the "case number" defined by the court. In reality, these should be considered charge numbers, because the values provided for each charge in a given case are typically related but unique. While the District Court Clerk's Manual (2021) recommends a common method for assigning case numbers ((case type)+year+(sequential number)+suffix), the Circuit Court Clerk's Manual (2020) acknowledges variations in numbering conventions across courts. For charges within each court, I first group charges into cases based on the criteria that charges are treated as a single case if they belong to the same defendant and the last 4 non-suffix digits of the case number are either identical or sequential. I then expand those groupings to include any additional charges that were filed against the same person on that same date— even if the case numbers appear unrelated.

In the included example, all six entries represent charges against the same individual. However, they are grouped as four distinct cases. The first three would be grouped together on either the case number or filing date criteria: because 4309, 4310, and 4311 are sequential these are treated as one case and they also were all filed on the same date. In contrast, none of the remaining charges show related case numbers or identical filing dates, so they are treated as separate— even though two of the charges were filed only two weeks apart.

For the analyses in which I omit probation violations and similar offenses, I exclude these charges before grouping the cases. For example, if an individual was sentenced to probation due to a charge on Jan 1, 2015 and then on Jan 1, 2016 was charged with violating that probation and another offense, they would appear in the data as having two separate cases, one stemming from the 2015 event and the other from the 2016 event.

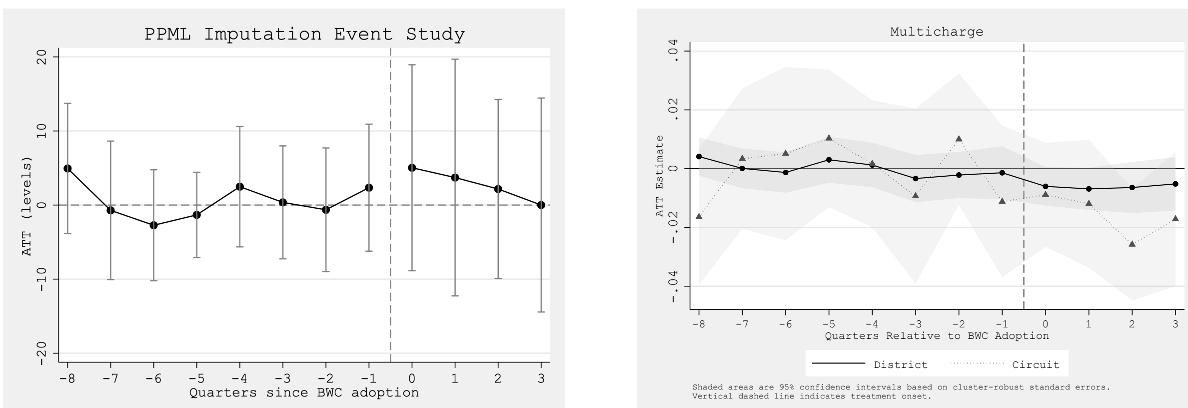
casenumber	fileddate	person_id	case_id_ba~s
[redacted]4310-00	21138	100000000000005	11
[redacted]4311-00	21138	100000000000005	11
[redacted]4309-00	21138	100000000000005	11
[redacted]8756-00	21319	100000000000005	13
[redacted]2897-00	21404	100000000000005	14
[redacted]3327-00	21418	100000000000005	15

B Event Study Plots



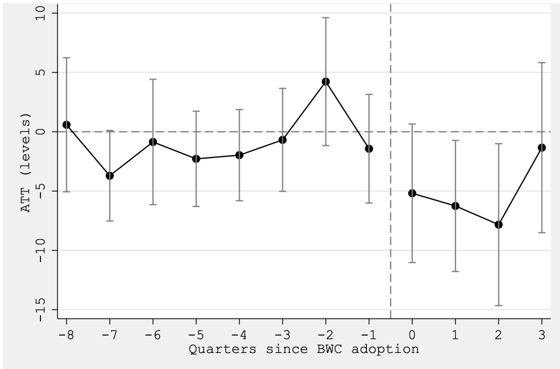
(a) (b)

Figure B.1: Event Studies: Case filing

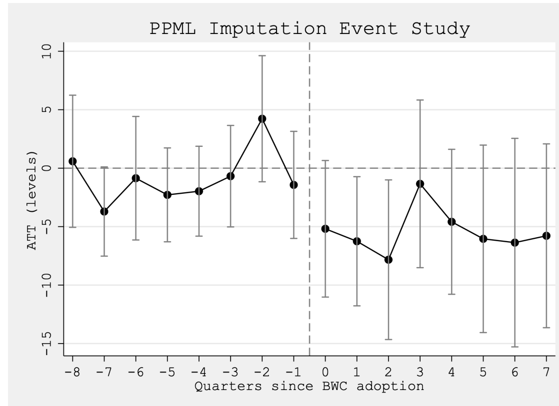


(a) (b)

Figure B.2: Event Studies: Appeals, Multi-charge

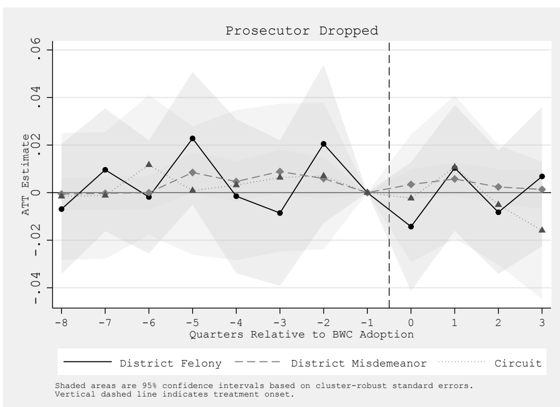


(a)

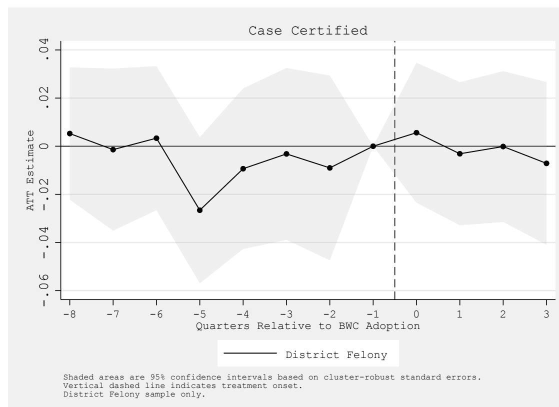


(b)

Figure B.3: Event Studies: Behav. Effect Cases (main, extended)

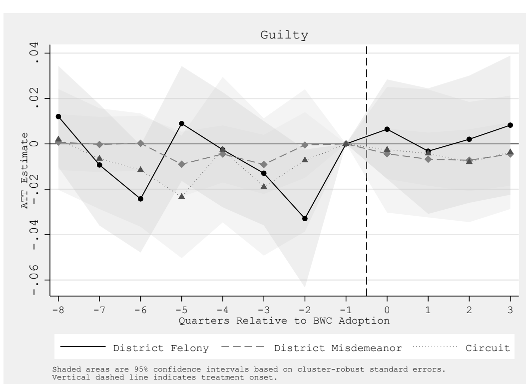


(a)

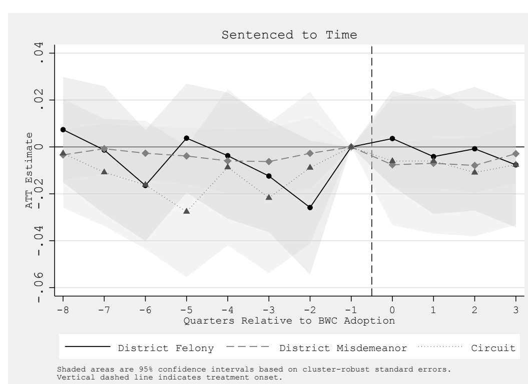


(b)

Figure B.4: Event Studies: Combined Dropped, Certified

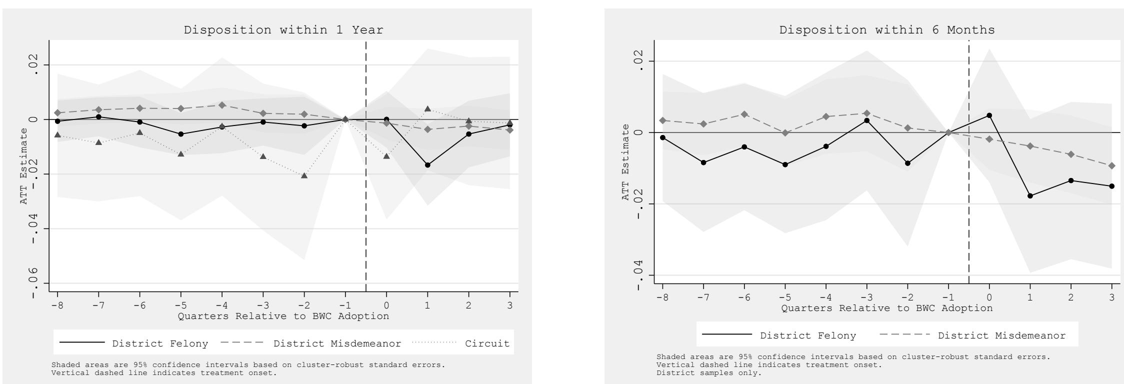


(a)



(b)

Figure B.5: Event Studies: Combined Guilty, Sentenced to Time



(a) (b)

Figure B.6: Event Studies: Combined 1 yr, 6 month Dispositions

C Main Table Robustness

Table C.1: District Court: 50% Threshold

Panel A: Filings

	Cases Filed	Behav Effect	Multicharge	Share Misd.	Share Infract.
Treatment Effect	-141.849	-4.296	-0.002	-0.005**	0.002
District	(334.255)	(2.811)	(0.004)	(0.002)	(0.006)
Mean	7434.345	53.380	0.185	0.929	0.713
N	3456	3456	3456	3456	3456

Panel B: Case Processes and Resolutions

	Prosecutor Dropped	Case Certified	Guilty	Sentenced to Time	Amend (N)	Disposition 1 Year	Disposition 6 Mos
District Felonies							
Treatment Effect	-0.002	-0.003	0.001	-0.005	-0.001	-0.007	-0.014
	(0.011)	(0.013)	(0.011)	(0.010)	(0.008)	(0.005)	(0.009)
Mean	0.390	0.572	0.297	0.247	0.150	0.964	0.874
N	3520	3520	3520	3520	3520	3520	3520
District Misdemeanors							
Treatment Effect	0.002	–	-0.004	-0.005	0.008	-0.003	-0.005
	(0.003)	–	(0.005)	(0.005)	(0.005)	(0.003)	(0.005)
Mean	0.093	–	0.749	0.191	0.199	0.956	0.887
N	3567	–	3567	3567	3567	3567	3567

Results from BJS imputation estimator. Cluster-robust standard errors in parentheses. Behavioral effect estimates analogously produced using poisson regression and boostrapped standard errors. Share of misdemeanors is presented relative to sum of misd. and felony cases, and is omitted in court-quarters with no misdemeanor/felony cases filed. Results from BJS imputation estimator. All estimates use population-weighted ATT. Panel A controls for local unemployment rate. Panel B Controls include average number of charges per case, shares of female and Black defendants, count of cases in the court, and the locality unemployment rate. * p<0.10, ** p<0.05, *** p<0.01. † indicates pre-trend test p<0.10.

Table C.2: Circuit Court: 50% threshold

Panel A: Filings

	Cases Filed (per 100k)	Appeals (per 100k)	Multicharge
Treatment Effect	-3.860	2.732	-0.012
	(9.182)	(4.390)	(0.007)
Mean	237.541	86.068	0.466
N	3183	3183	3176

Panel B: Case Processes and Resolutions

	Prosecutor Dropped	Guilty	Sentenced to Time	Amend (N)	Disposition 1 Year
Treatment Effect	-0.008	-0.007	-0.010	-0.006	-0.005
	(0.011)	(0.010)	(0.011)	(0.007)	(0.009)
Mean	0.263	0.728	0.721	0.132	0.821
N	3162	3162	3162	3162	3162

Results from BJS imputation estimator. Cluster-robust standard errors in parentheses. Behavioral effect estimates analogously produced using poisson regression and bootstrapped standard errors. Panel A controls for local unemployment rate. Panel B Controls include average number of charges per case, shares of female and Black defendants, count of cases in the court, and the locality unemployment rate. * $p<0.10$, ** $p<0.05$, *** $p<0.01$. † indicates pre-trend test $p<0.10$.

Table C.3: Case Filing Robustness Across Estimation Methods

	District	Circuit	District	Circuit	District	Circuit	District	
	Cases Filed pc	Cases Filed pc	Beh. Cases pc	Appeals pc	Multi-charge	Multi-charge	Share Misd.	Share Infract.
TWFE	-146.801	-6.591	-5.761**	-0.000	-0.006	-0.015*	-0.005*	0.005
BJS Alt Controls	-135.327	-4.200	-1.913	0.000	-0.006	-0.018	-0.002	-0.004
BJS Only Adopters	60.346	3.736	-3.689	0.000	-0.009***	-0.016	-0.003	0.001
JWDID	41.442	3.569	-3.869	0.000	-0.014**	-0.016	0.001	0.002
Wooldridge DID	-141.348	-4.215	-1.934	0.000	-0.006	-0.018	-0.002	-0.004
CSDID	354.036	-8.186	-4.043*	-0.000	-0.005	-0.007	-0.002	0.002
Main Extended	-340.772	-12.063	-7.148	-0.000	-0.006	-0.020	-0.005	0.005
Main Unbalanced	-52.020	-7.497	-6.873***	-0.000	-0.002	-0.020*	-0.002	0.005
Main	-144.188	-5.852	-5.790**	-0.000	-0.006*	-0.016**	-0.005**	0.005
Main Poisson	–	–	-5.152*	1.204	–	–	–	–
Mean	7451.271	238.664	52.855	0.001	0.185	0.463	0.929	0.711

Notes: *** p<0.01, ** p<0.05, * p<0.10. Main shows primary BJS specification for all outcomes except behavioral effect cases and appeals, for which the primary specification poisson is showed in the Main Poisson row, and OLS is presented in line. Only Adopters restricts treated group to units treated before 2018, and uses later-treated units as controls. Additional controls for the Alt Controls specification include population and unit-specific trends. Main unbalanced includes later-treated units that do not have observations in all four post-periods.

Table C.4: Process and Resolution Robustness Checks Across Estimation Methods

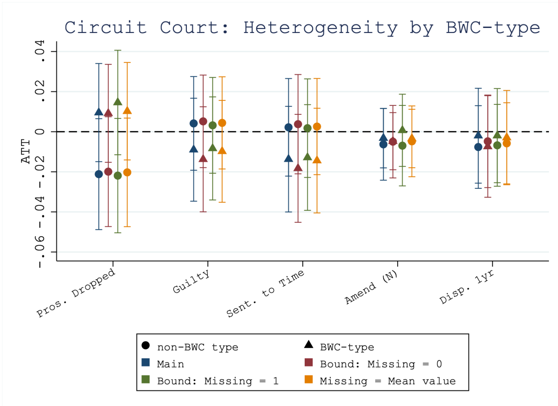
	Dropped	Certified	Guilty	Sentenced	Amend	Disp 1yr	Disp 6mo
District Felony Cases							
TWFE	-0.001	-0.000	0.004	-0.002	0.003	-0.006	-0.011
Stacked	-0.008	-0.000	0.006	0.002	0.000	-0.002	0.001
BJS Alt Controls	0.003	-0.006	0.020	0.013	0.009	-0.007	-0.010
BJS Only Adopters	0.002	-0.003	0.002	-0.000	0.006	-0.003	0.003
JWDID	-0.004	0.005	0.004	0.001	0.001	-0.005	-0.004
Wooldridge DID	-0.001	-0.007	0.017	0.011	0.009	-0.007	-0.010
CSDID	-0.052	0.011	0.012	0.013	0.015	-0.004	-0.006
Main Extended	-0.005	-0.001	0.011	0.000	0.000	-0.009	-0.012
Main Unbalanced	-0.001	-0.001	0.005	-0.004	0.002	-0.010*	-0.011
Main	-0.001	-0.002	0.004	-0.002	0.002	-0.006	-0.011
Mean	0.391	0.571	0.302	0.250	0.150	0.966	0.880
District Misd Cases							
TWFE	0.003	–	-0.005	-0.007	0.006	-0.003	-0.005
Stacked	-0.002	–	-0.002	-0.008*	0.001	-0.002	-0.001
BJS Alt Controls	-0.001	–	-0.003	-0.002	0.004	-0.005*	-0.006
BJS Only Adopters	0.000	–	-0.008	-0.011*	-0.000	-0.009***	-0.010*
JWDID	0.001	–	-0.008	-0.013*	0.000	-0.013***	-0.015**
Wooldridge DID	-0.002	–	-0.003	-0.003	0.005	-0.005	-0.006
CSDID	-0.001	–	-0.006	-0.008	0.004	-0.002	-0.004
Main Extended	0.001	–	-0.002	-0.008	0.008	-0.002	-0.006
Main Unbalanced	-0.002	–	-0.004	-0.007	0.003	-0.000	-0.005
Main	0.003	–	-0.006	-0.007	0.007	-0.003	-0.005
Mean	0.093	–	0.750	0.193	0.197	0.957	0.890
Circuit Felony Cases							
TWFE	-0.002	–	-0.003	-0.006	-0.003	-0.002	–
Stacked	0.006	–	0.016	0.017	0.004	0.017	–
BJS Alt Controls	-0.008	–	0.005	0.009	-0.001	0.005	–
BJS Only Adopters	-0.016	–	-0.001	-0.001	-0.006	-0.000	–
JWDID	-0.015	–	-0.001	-0.001	-0.006	-0.002	–
Wooldridge DID	-0.008	–	0.005	0.009	-0.001	0.005	–
CSDID	-0.037**	–	0.013	0.015	-0.009	0.017	–
Main Extended	-0.010	–	-0.011	-0.013	-0.003	-0.012	–
Main Unbalanced	-0.017	–	-0.004	-0.016	0.008	-0.005	–
Main	-0.003	–	-0.005	-0.008	-0.004	-0.003	–
Mean	0.261	–	0.728	0.721	0.132	0.821	–

*** p<0.01, ** p<0.05, * p<0.10. Main includes primary BJS specification for all outcomes. Only Adopters restricts treated group to units treated before 2018, and uses later-treated units as controls. Additional controls for the Alt Controls specification include court-specific time trends (all) and supplemental charge-class categories and a multi-charge indicator (district).

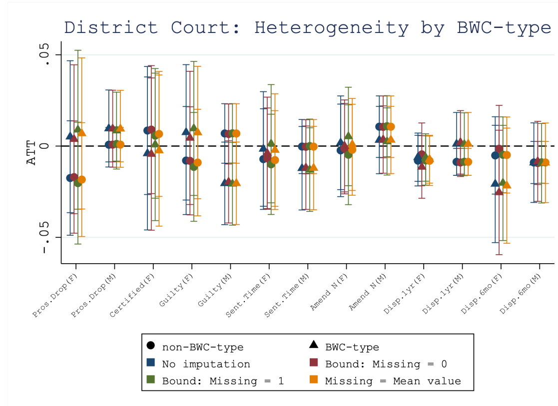
Table C.5: Circuit Court Process/Resolution: No 1 yr Condition

	Prosecutor Dropped	Guilty	Sentenced to Time	Amend (N)	Disposition 1 Year
Treatment Effect	-0.004	0.001	-0.001	0.000	–
	(0.011)	(0.007)	(0.008)	(0.007)	–
Mean	0.290	0.836	0.827	0.162	–
N	3062	3062	3062	3062	–

Results from BJS imputation estimator. Cluster-robust standard errors in parentheses. This specification omits the requirement that cases resolve within one year of filing. Controls include average number of charges per case, shares of female and Black defendants, count of cases in the court, and the locality unemployment rate. * $p<0.10$, ** $p<0.05$, *** $p<0.01$. † indicates pre-trend test $p<0.10$.

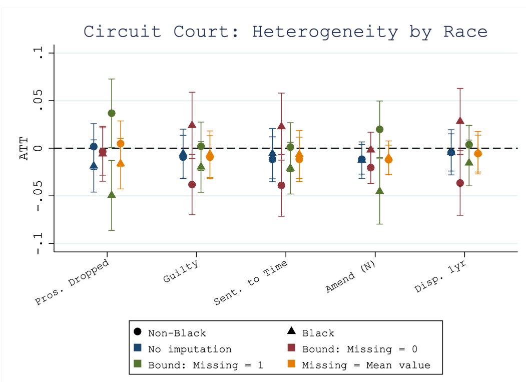


(a)

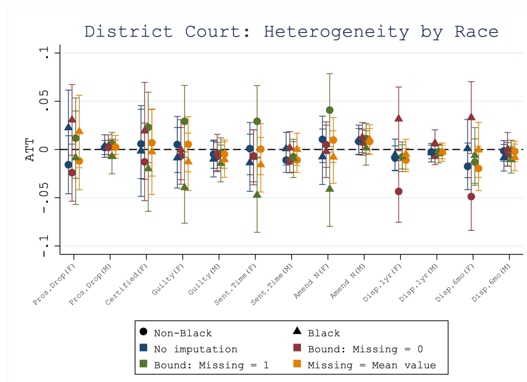


(b)

Figure C.1: Bounds on case type heterogeneity analyses constructed by imputing missing values



(a)



(b)

Figure C.2: Bounds on racial heterogeneity analyses constructed by imputing missing values

D Main Table Supplemental Analyses

Table D.1: District Court: Extended frame (2-year post-period)

Panel A: Filings							
	Cases Filed	Behav Effect	Multicharge	Share Misd.	Share Infract.		
Treatment Effect	-340.772	-6.608**	-0.006	-0.005**	0.005		
District	(430.633)	(3.290)	(0.004)	(0.003)	(0.006)		
Mean	7614.703	52.696	0.184	0.930	0.714		
N	3357	3357	3357	3357	3357		
Panel B: Processes and Resolutions							
	Prosecutor Dropped	Case Certified	Guilty	Sentenced to Time	Amend	Disposition 1 Year	Disposition 6 Mos
Treatment Effect	-0.006	0.000	0.011	0.000	0.000	-0.009	-0.012
District (Fel.)	(0.012)	(0.014)	(0.013)	(0.012)	(0.008)	(0.006)	(0.011)
Mean	0.385	0.578	0.296	0.244	0.147	0.965	0.880
N	3307	3307	3307	3307	3307	3307	3307
Treatment Effect	0.001	–	-0.002	-0.008	0.007	-0.002	-0.005
District (Misd.)	(0.003)	–	(0.006)	(0.005)	(0.007)	(0.004)	(0.006)
Mean	0.092	–	0.750	0.191	0.198	0.956	0.888
N	3357	–	3357	3357	3357	3357	3357

Results from BJS imputation estimator for an extended, 8-post period sample. Cluster-robust standard errors in parentheses. Behavioral effect estimates analogously produced using poisson regression and boostrapped standard errors. Share of misdemeanors is presented relative to sum of misd. and felony cases, and is omitted in court-quarters with no misdemeanor/felony cases filed. Results from BJS imputation estimator. All estimates use population-weighted ATT. Panel A controls for locality unemployment rate. Panel B additionally controls for charges per case, race and gender shares, total number of cases in the sample for the court-quarter. * p<0.10, ** p<0.05, *** p<0.01. † indicates pre-trend test p<0.10.

Table D.2: Circuit Court: Extended frame (2-year post-period)

Panel A: Circuit Court Filings					
	Cases Filed (per 100k)	Appeals (per 100k)	Multicharge		
Treatment Effect	-12.063	-0.530	-0.020**		
	(10.104)	(4.380)	(0.009)		
Mean	239.775	85.756	0.466		
N	3076	3076	3069		
Panel B: Circuit Court Case Outcomes					
	Prosecutor Dropped	Guilty	Sentenced to Time	Amend (N)	Disposition 1 Year
Treatment Effect	-0.010	-0.011	-0.013	-0.003	-0.012
	(0.011)	(0.011)	(0.012)	(0.006)	(0.010)
Mean	0.265	0.728	0.720	0.131	0.820
N	3057	3057	3057	3057	3057

Results from BJS imputation estimator with extended balanced eight post-period sample. Cluster-robust standard errors in parentheses. Behavioral effect estimates analogously produced using poisson regression and bootstrapped standard errors. Panel A controls for local unemployment rate. Panel B Controls include average number of charges per case, shares of female and Black defendants, count of cases in the court, and the locality unemployment rate. * p<0.10, ** p<0.05, *** p<0.01. † indicates pre-trend test p<0.10.

Table D.3: District Court: Population Weighted Estimates

Panel A: Filings							
	Cases Filed (per 100k)	Behav Effect (per 100k)	Multicharge	Share Misd.	Share Infract.		
Treatment Effect	-87.559	–	-0.002	-0.007***	0.000		
SE	(186.020)	–	(0.004)	(0.002)	(0.007)		
Panel B: Case Processes and Resolutions							
	Pros. Dropped	Certified	Guilty	Sent. to Time	Amend (N)	Disp. 1 Yr	Disp. 6 Mos
Felony							
Treatment Effect	-0.009	0.001	0.007	-0.003	-0.002	-0.023	-0.035**
SE	(0.012)	(0.014)	(0.013)	(0.012)	(0.007)	(0.015)	(0.016)
Misdemeanor							
Treatment Effect	0.000	–	-0.009	-0.017***	0.005	-0.008	-0.011*
SE	(0.003)	–	(0.007)	(0.006)	(0.006)	(0.007)	(0.007)

Results from BJS imputation estimator. All estimates use population-weighted ATT. Panel A controls for unemployment rate. Panel B controls for locality unemployment rate, charges per case, share of female and Black defendants. Cluster-robust standard errors in parentheses. * $p<0.10$, ** $p<0.05$, *** $p<0.01$. † indicates pre-trend test $p<0.10$.

Table D.4: Circuit: Population Weighted Estimates

Panel A: Filings					
	Cases Filed (per 100k)	Appeals (per 100k)	Multicharge		
Treatment Effect	-26.569***	-8.491**	-0.011		
SE	(6.572)	(4.308)	(0.013)		
Panel B: Processes and Resolutions					
	Prosecutor Dropped	Guilty	Sentenced to Time	Amend	Disposition 1 Year
Treatment Effect	-0.005	-0.008	-0.013	-0.002	-0.004
SE	(0.011)	(0.010)	(0.010)	(0.007)	(0.010)
-					

Results from BJS imputation estimator. All estimates use population-weighted ATT. Panel A controls for local unemployment rate. Panel B controls include average number of charges per case, shares of female and Black defendants, count of cases in the court, and the locality unemployment rate. Cluster-robust standard errors in parentheses. * $p<0.10$, ** $p<0.05$, *** $p<0.01$. † indicates pre-trend test $p<0.10$.

Figure D.1: Behavioral Effect Cases: Court ATTs by Population, Adoption Cohort

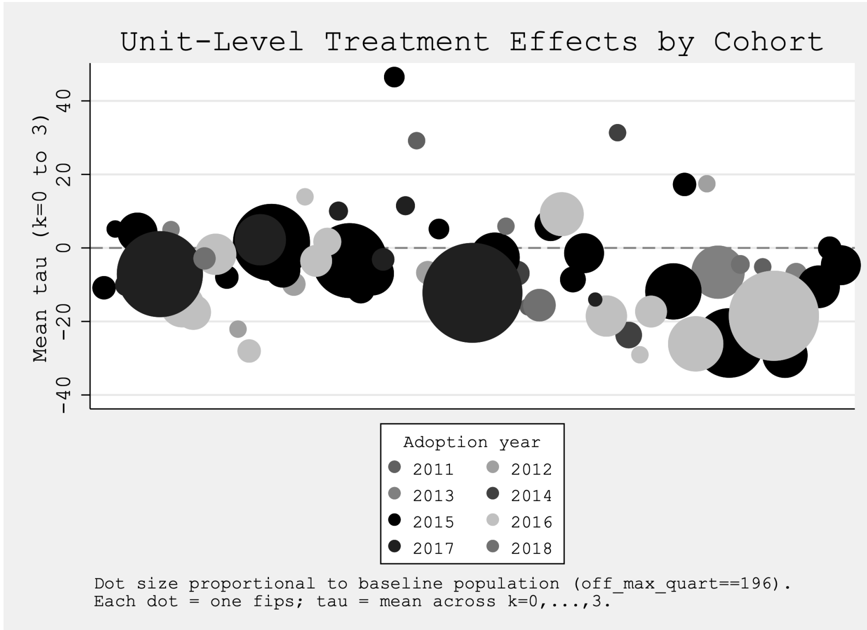


Figure D.2: Appeals: Court ATTs by Population

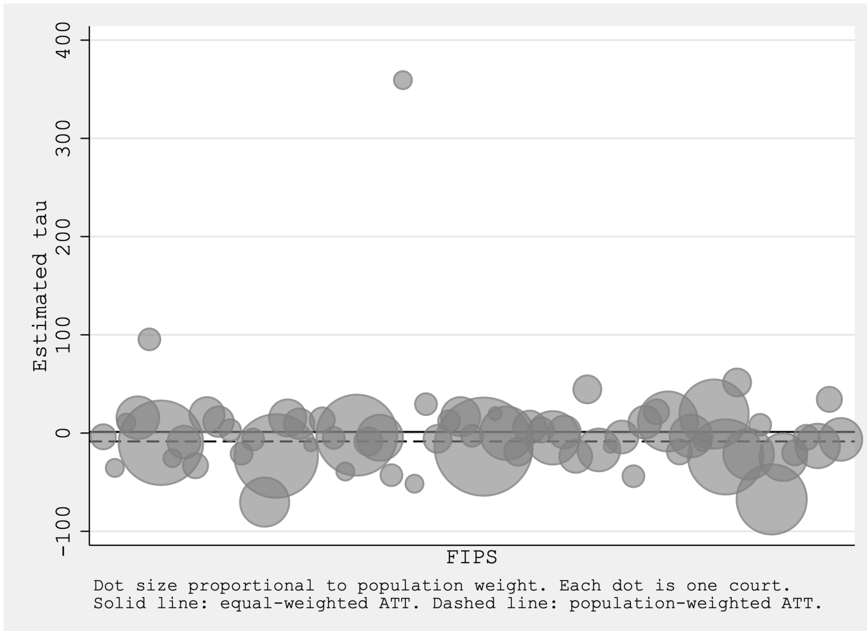


Figure D.3: Circuit Filings: Court ATTs by Population

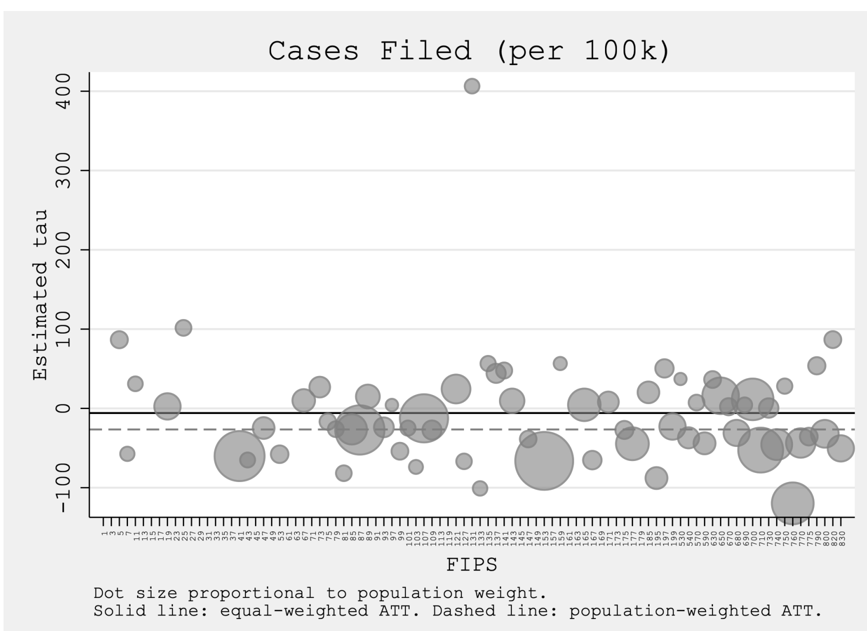


Table D.5: Processes and Resolutions per 100k residents

Panel A: District Court							
	Prosecutor Dropped	Case Certified	Guilty	Sentenced to Time	Amend (N)	Disposition 1 Year	Disposition 6 Mos
Treatment Effect	1.311	1.346	-1.121	-0.799	0.281	2.677	0.280
District (Fel.)	(2.620)	(3.416)	(2.297)	(2.175)	(1.499)	(4.405)	(3.727)
Mean	70.659	103.175	54.518	45.694	26.606	175.177	158.848
N	3306	3306	3306	3306	3306	3306	3306
Treatment Effect	5.042	–	16.076	-23.942***	66.371	9.395	6.551
District (Misd.)	(6.096)	–	(97.263)	(8.536)	(52.617)	(100.562)	(97.641)
Mean	132.474	–	1599.077	286.883	476.384	1917.380	1802.295
N	3353	–	3353	3353	3353	3353	3353
Panel B: Circuit Court							
	Prosecutor Dropped	Guilty	Sentenced to Time	Amend	Disposition 1 Year		
Treatment Effect	0.967	–	0.656	0.418	-0.267	0.153	–
	(2.441)	–	(2.973)	(3.013)	(1.105)	(3.346)	–
Mean	44.454	–	119.537	118.304	21.334	135.666	–
Treatment Effect	1.359†	–	3.481	3.252	0.904	0.153	–
No 1-yr cond.	(2.656)	–	(3.410)	(3.439)	(1.285)	(3.346)	–
Mean	44.454	–	119.537	118.304	21.334	135.666	–
N	3062	–	3062	3062	3062	3062	–

Results from BJS imputation estimator. Cluster-robust standard errors in parentheses. Panel A controls for local unemployment rate. Panel B Controls include average number of charges per case, shares of female and Black defendants, count of cases in the court, and the locality unemployment rate. * p<0.10, ** p<0.05, *** p<0.01. † indicates pre-trend test p<0.10.

E Back of Envelope Caseload Calculations

In the body of the paper I reference a data point showing the average caseload for Virginia indigent defenders before body-worn camera adoption was 320 cases and cite this as evidence toward attorneys facing binding time constraints. A simple back-of-the-envelope calculation shows why this is the case. I show in Figure D.1 an attorney's production possibilities frontier under ABA guidelines, as well as the possible combinations of felony and misdemeanor cases that an attorney can take to total 320. Attorneys representing 320 cases can do so while adhering to ABA guidelines if their case combination lies on or under the ABA Guidelines curve (shown in green). This will only happen if they represent 48 or fewer felonies (15 percent of their caseload). However, the same report shows that felonies comprise over 30 percent of the cases overall, and a 3:4 ratio of felonies to misdemeanors when case types such as parole violations are excluded. Thus, it is reasonable to conclude that the caseloads faced by public defenders in Virginia before body-worn camera adoption lie outside the ABA production possibilities frontier and so indicate a binding time constraint under the ABA guidelines.

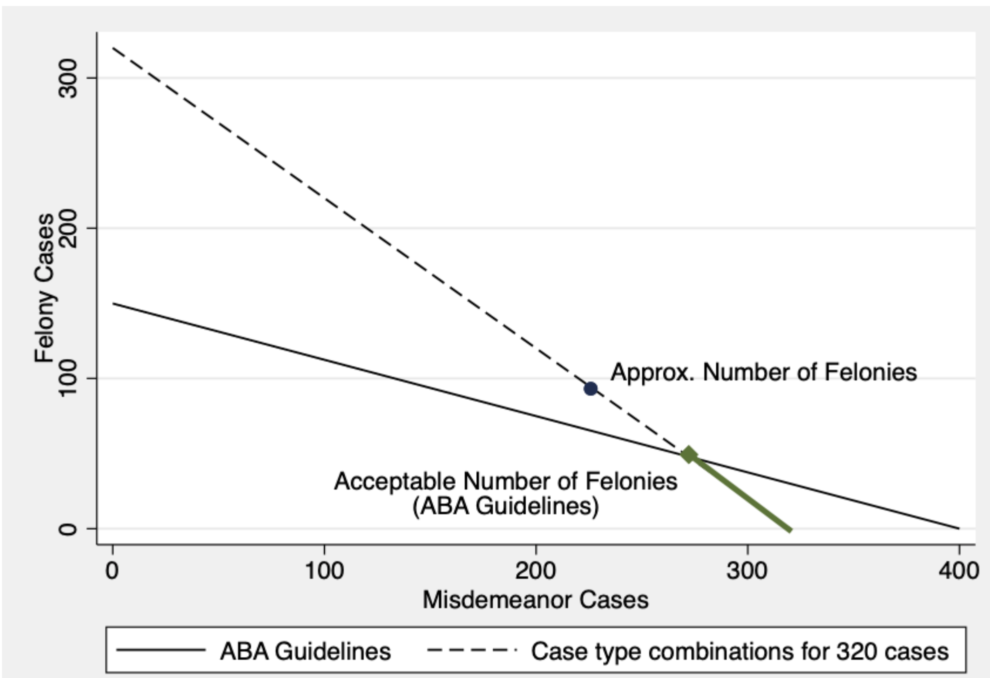


Figure D.1: PPF for defense attorneys under ABA guidelines and caseloads for VA public defenders